



Topics for Today

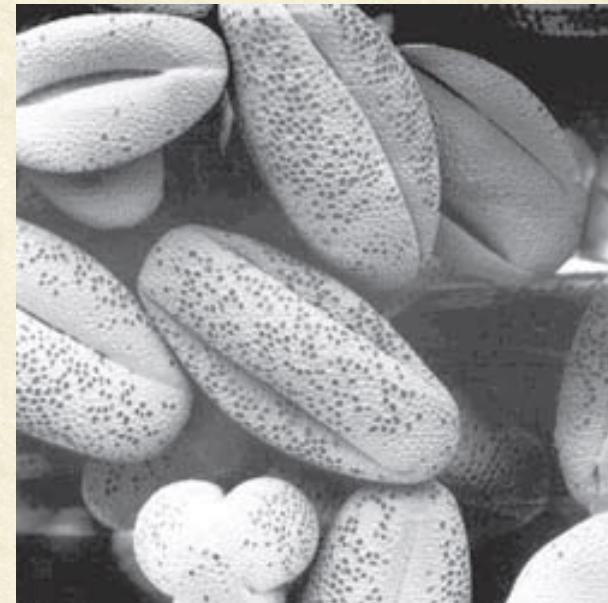
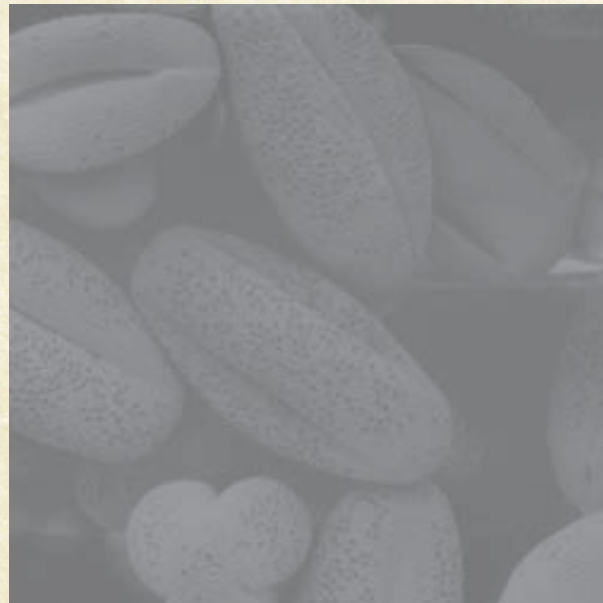
Histogram Equalization: Recap

- I. Histogram Processing: Continued
- II. Spatial vs. Transform Domain Processing
- III. Spatial Domain Processing
- IV. Spatial Filtering



CS7.404: Digital Image Processing

Monsoon 2025: **0**: Pixel Processing
(Histogram Manipulation: Recap)



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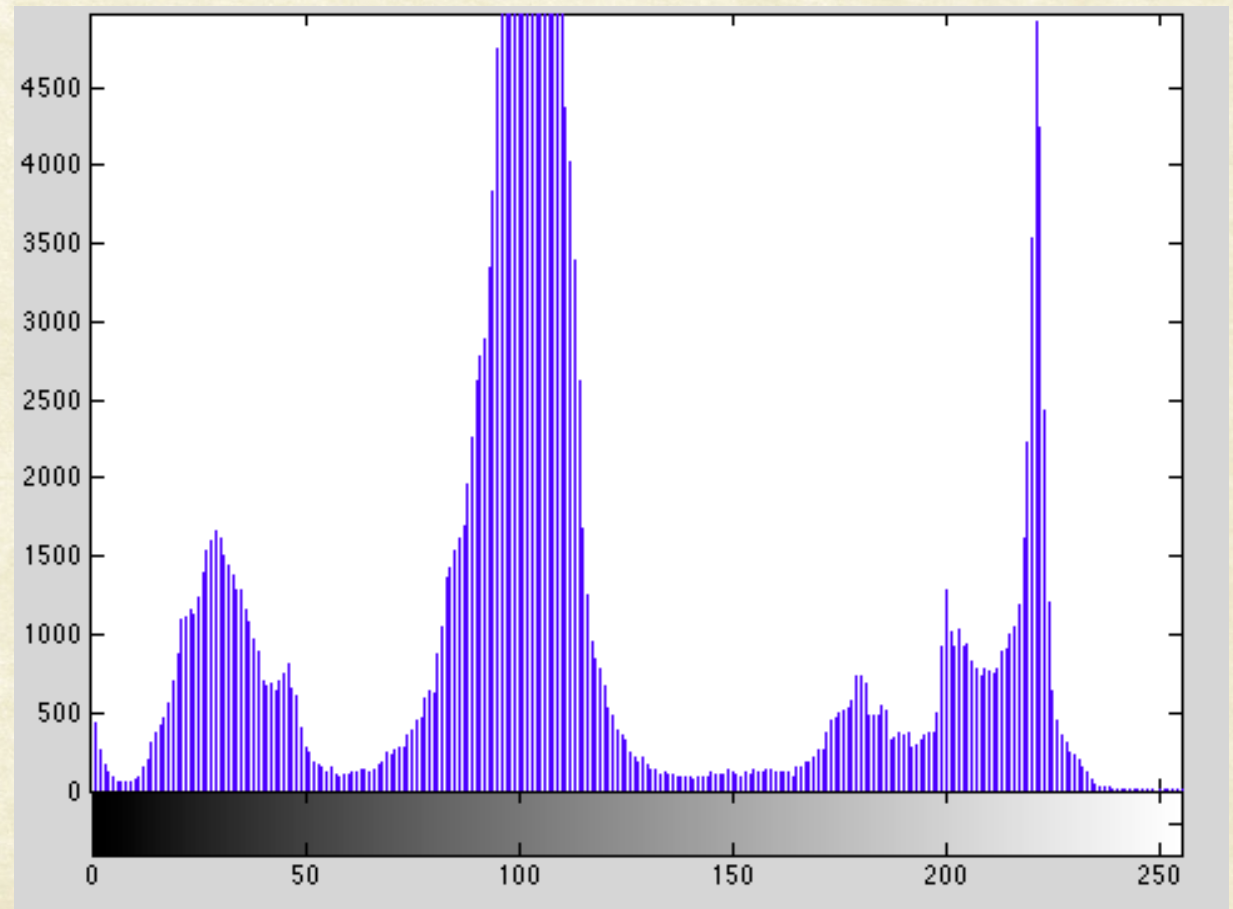


Histogram

$$h_r(i) = n_i$$

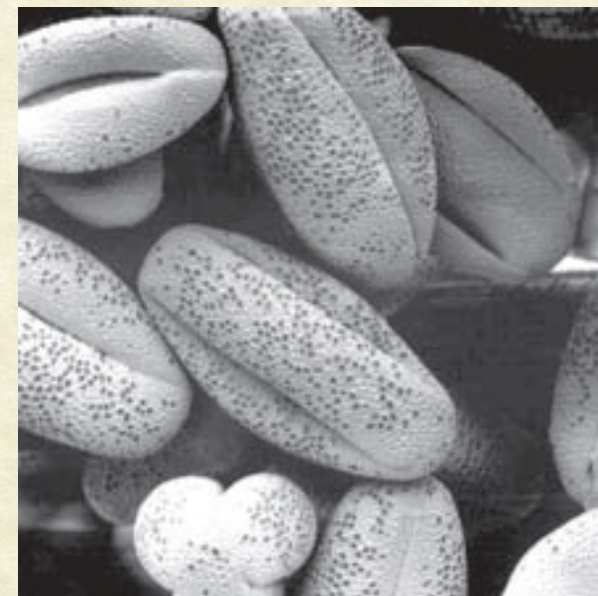
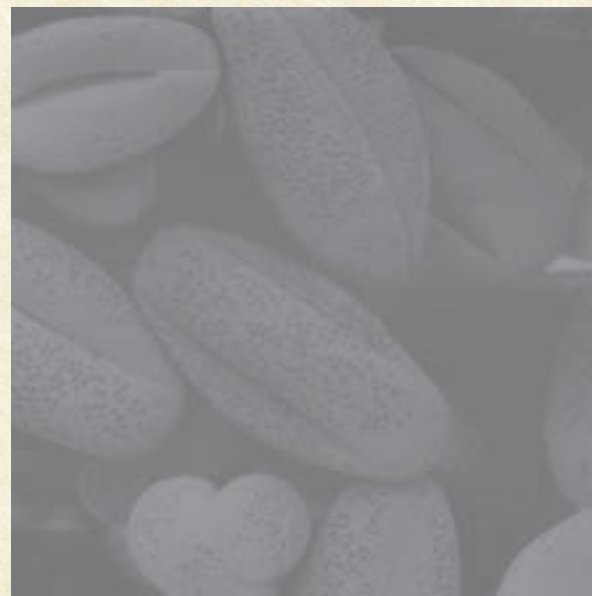
$i \rightarrow$ intensity value, range $[0, L-1]$

$n_i \rightarrow$ number of pixels with intensity i

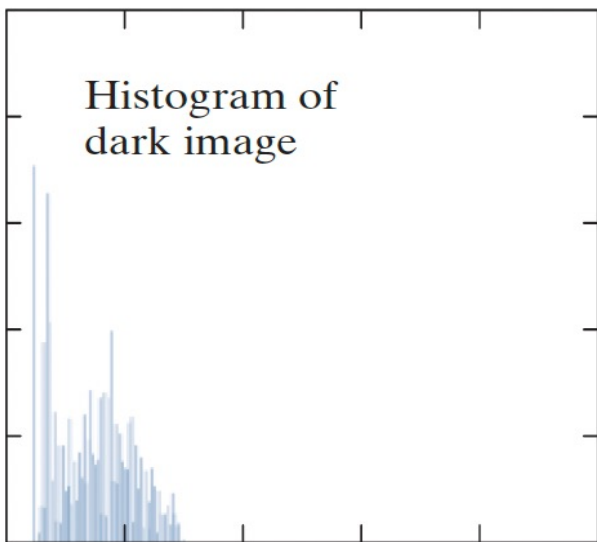




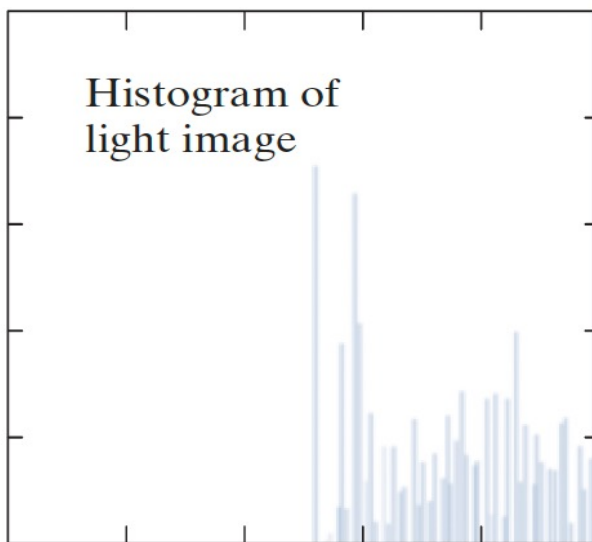
Histogram and Contrast



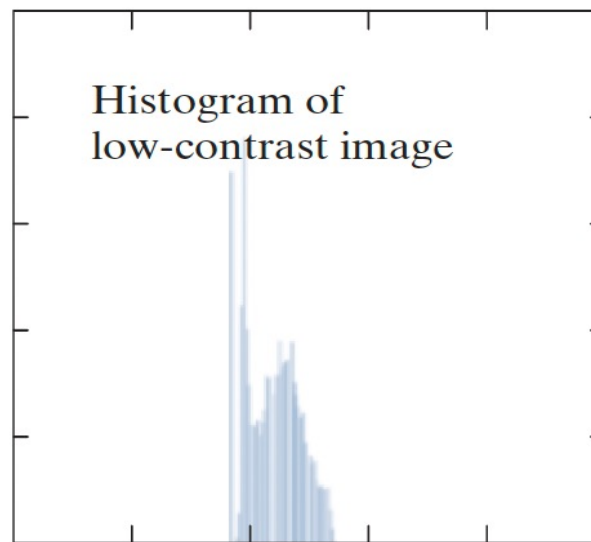
Histogram of
dark image



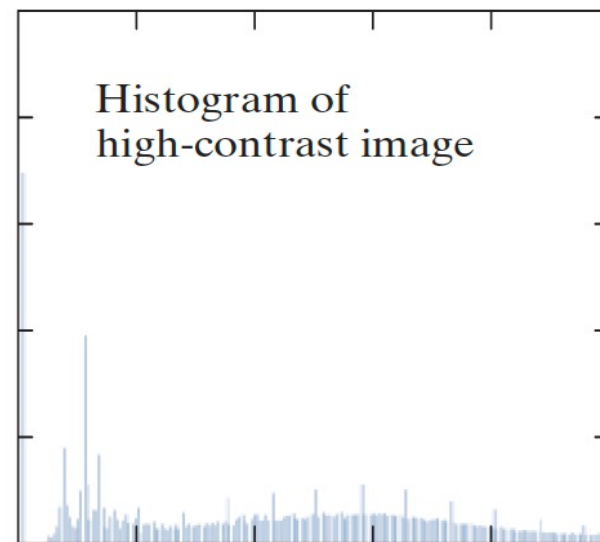
Histogram of
light image



Histogram of
low-contrast image

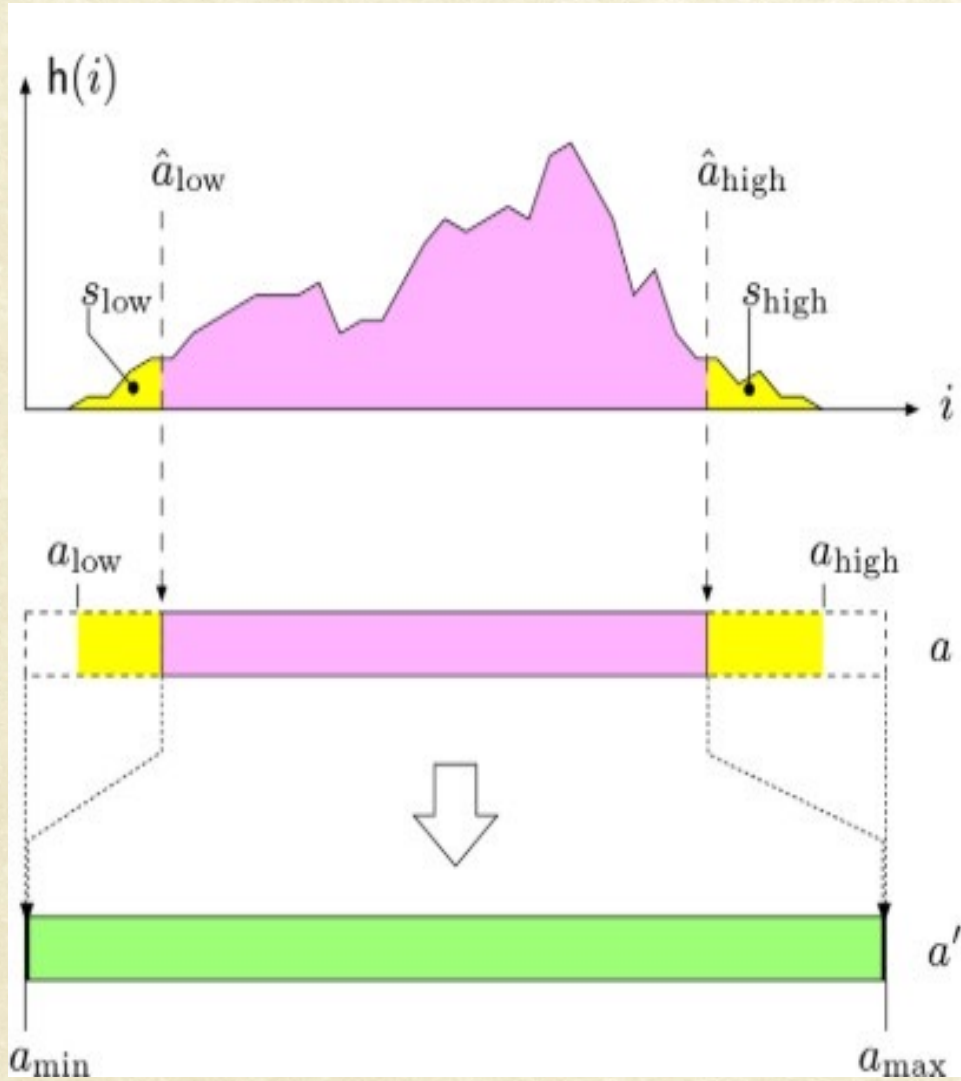


Histogram of
high-contrast image





Contrast Stretching: Ver.-2



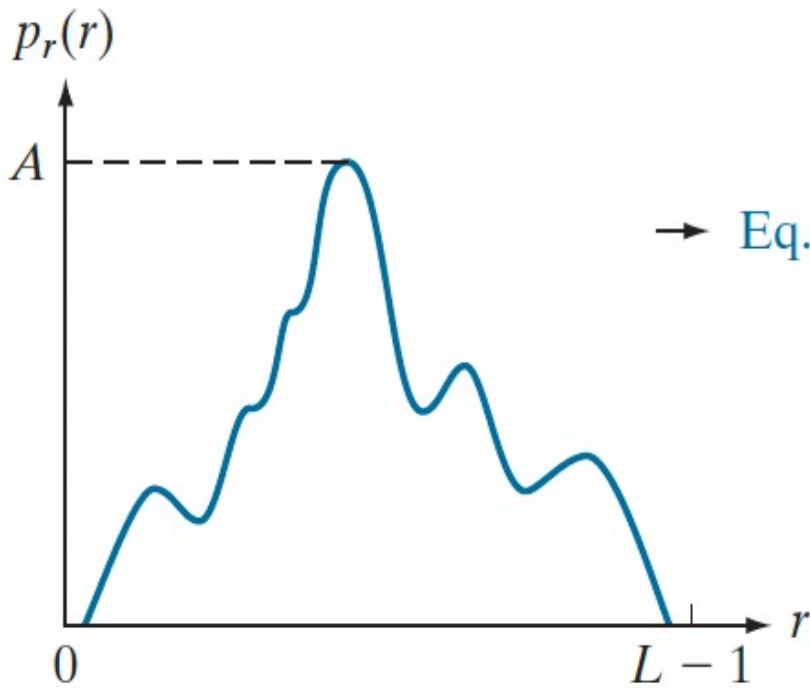
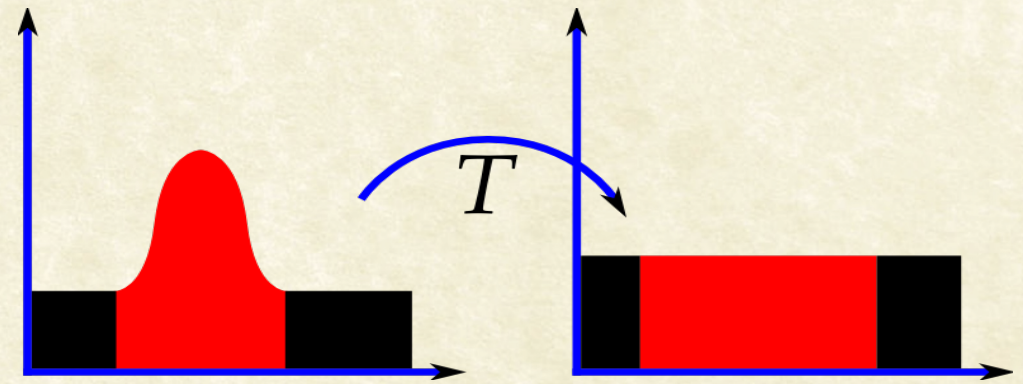
$$\hat{a}_{\text{low}} = \min\{i \mid H(i) \geq M \cdot N \cdot s_{\text{low}}\}$$

$$\hat{a}_{\text{high}} = \max\{i \mid H(i) \leq M \cdot N \cdot (1 - s_{\text{high}})\}$$

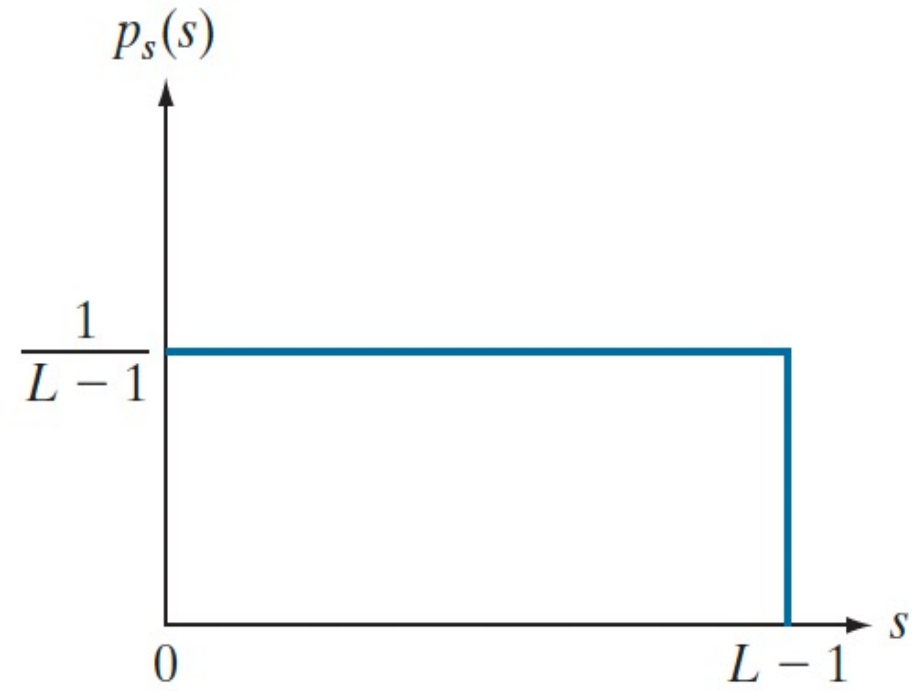
$$f_{\text{mac}}(a) = \begin{cases} a_{\text{min}} & \text{for } a \leq \hat{a}_{\text{low}} \\ a_{\text{min}} + (a - \hat{a}_{\text{low}}) \cdot \frac{a_{\text{max}} - a_{\text{min}}}{\hat{a}_{\text{high}} - \hat{a}_{\text{low}}} & \text{for } \hat{a}_{\text{low}} < a < \hat{a}_{\text{high}} \\ a_{\text{max}} & \text{for } a \geq \hat{a}_{\text{high}} \end{cases}$$



Histogram Equalization



→ Eq. (3-11) →

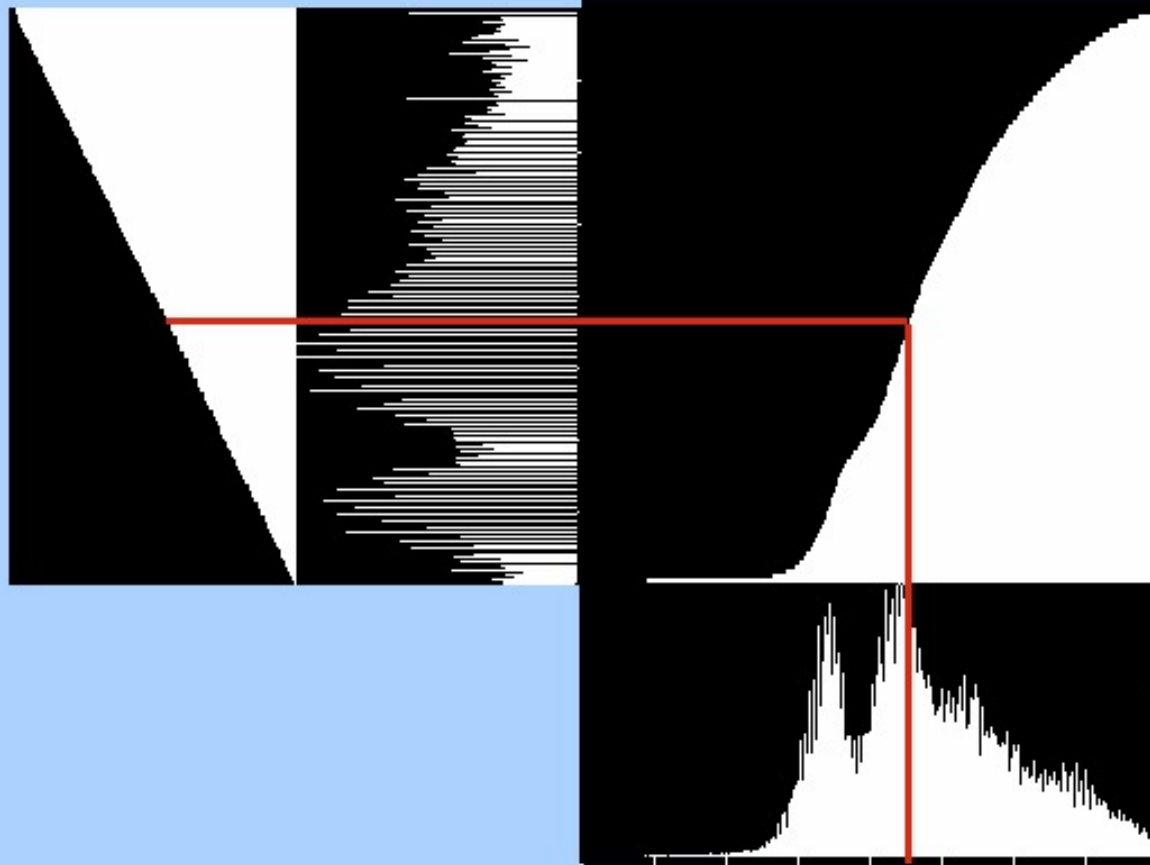




Computing the Intensity-Mapping



Equalized histogram



Histogram of original image

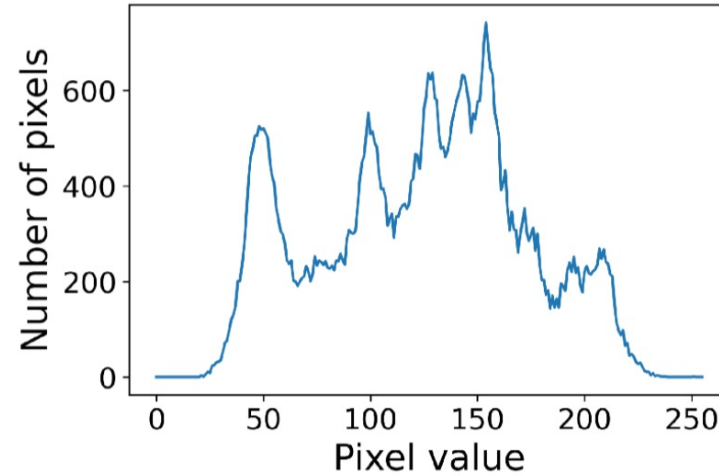




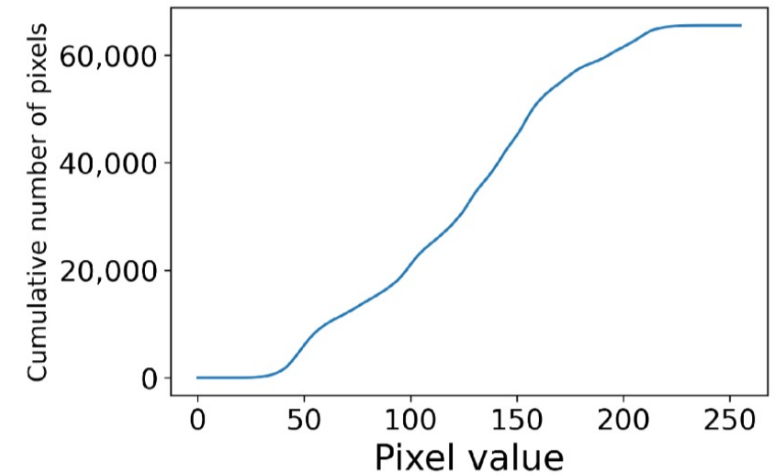
Computing the Mapping for Histogram Eq.



(a) Original image



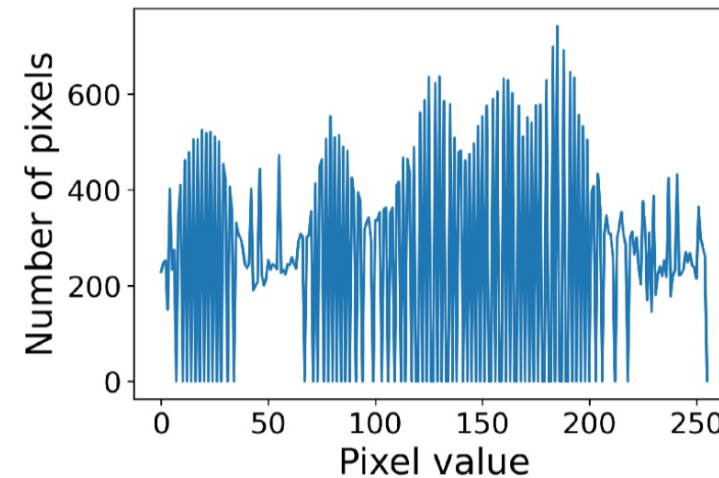
(b) Histogram



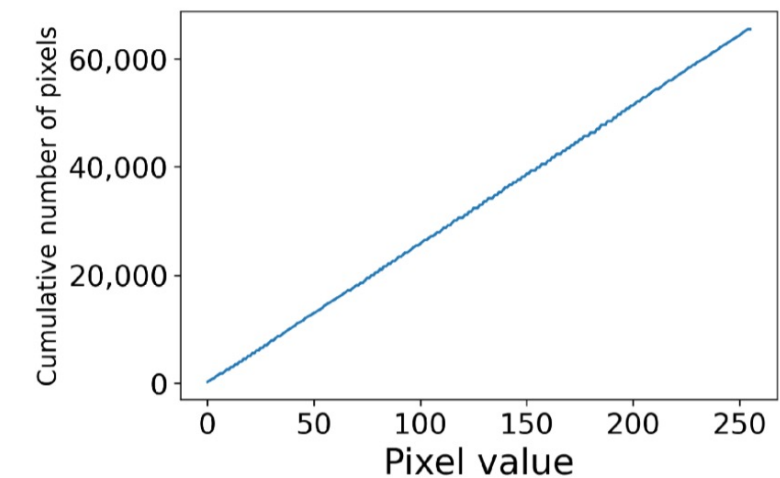
(c) Cumulative histogram



(d) Histogram-equalized image



(e) Histogram



(f) Cumulative histogram



Histogram Equalization

$$p_r(r_k) = \frac{n_k}{MN}$$

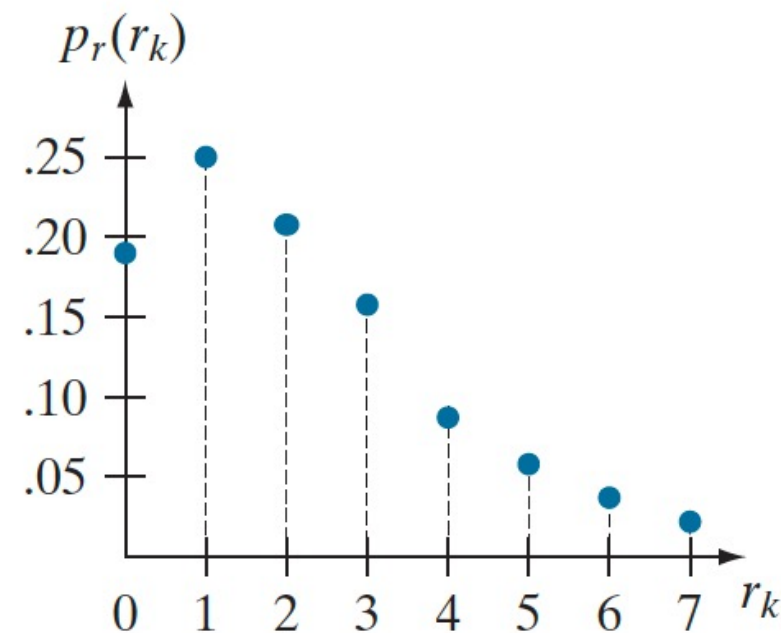
$$s_k = T(r_k) = \text{round} \left[(L - 1) \sum_{j=0}^k p_r(r_j) \right]; \quad k = 0, 1, 2, \dots, L - 1$$



Histogram Equalization - Example

r_k	n_k	$p_r(r_k) = n_k / MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

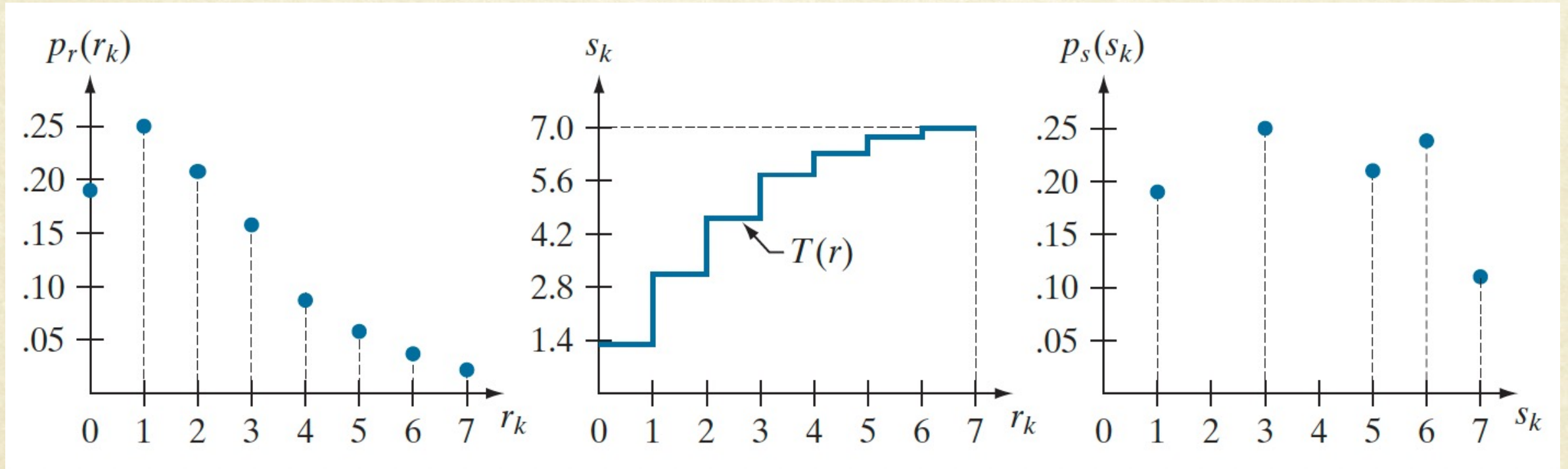
Intensity distribution and histogram values for a 3-bit, 64 x 64 digital image.





Histogram Equalization - Example

$$s_k = \text{round} \left[(L - 1) \sum_{j=0}^k p_r(r_k) \right]$$



(a) Original histogram;

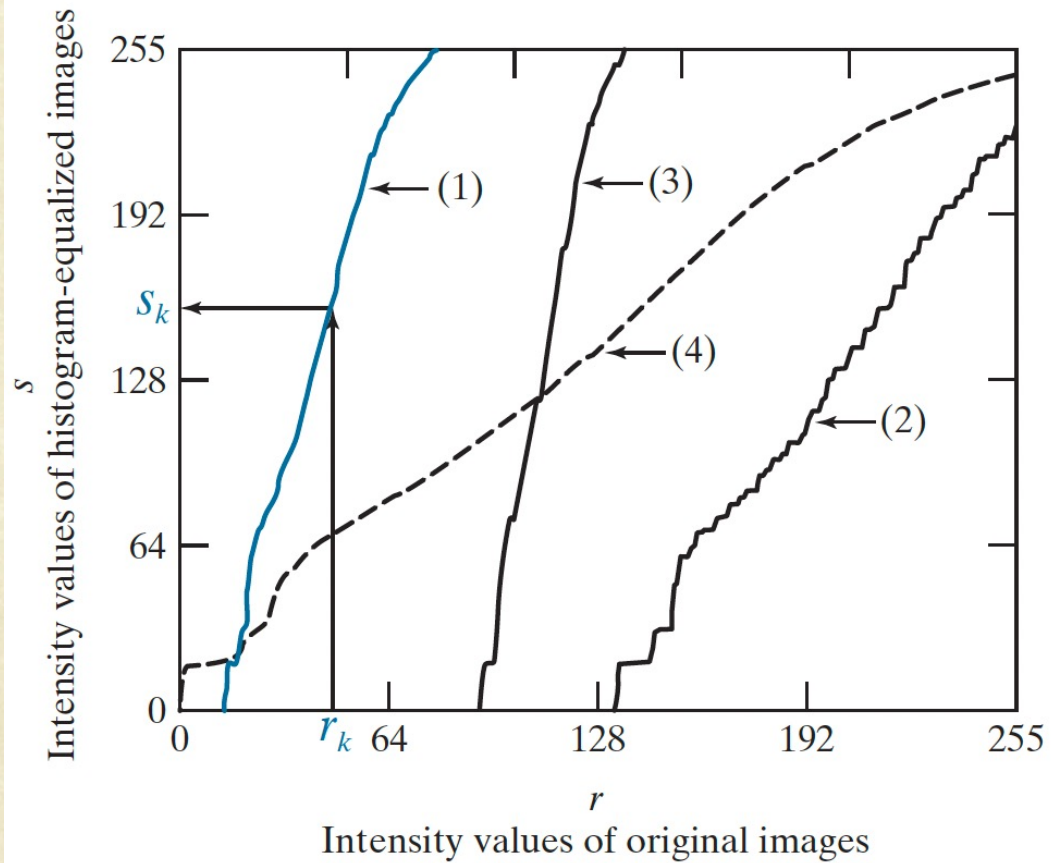
(b) Transformation function;

(c) Equalized histogram.

FIGURE 3.19: Histogram equalization.



Histogram Equalization



1



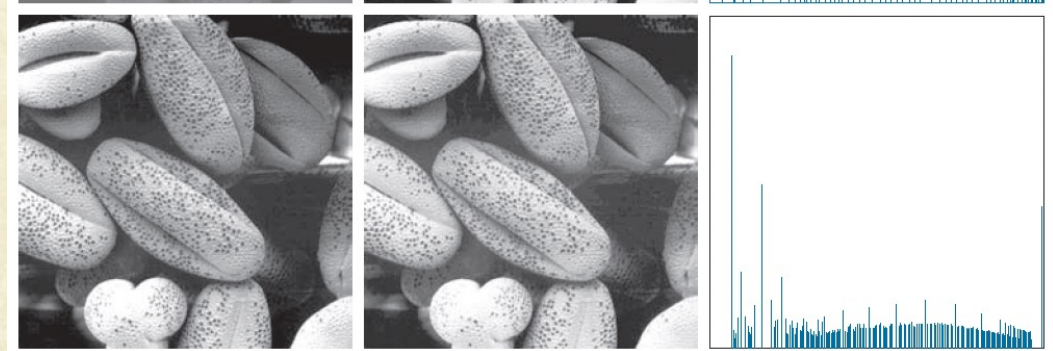
2



3



4





Histogram Equalization v/s Contrast Enhancement



Contrast Enhancement

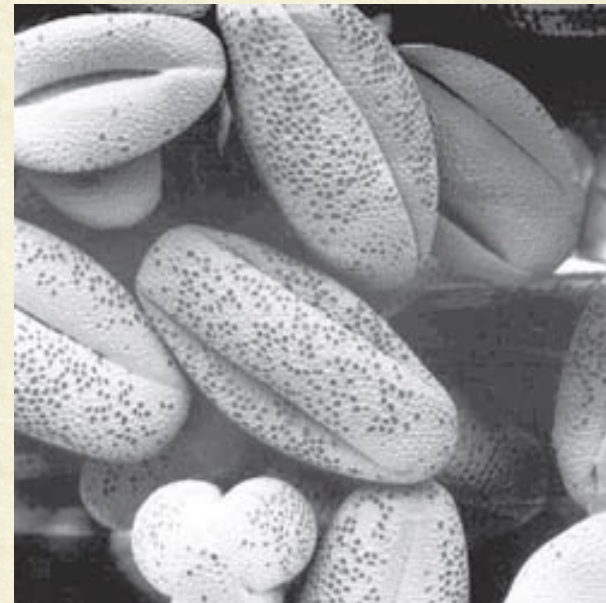
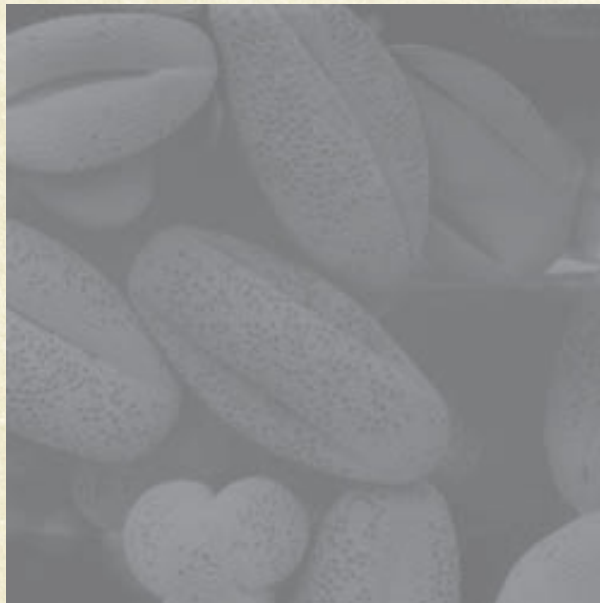


Histogram equalization



CS7.404: Digital Image Processing

Monsoon 2025: **I**: Histogram Processing: Continued



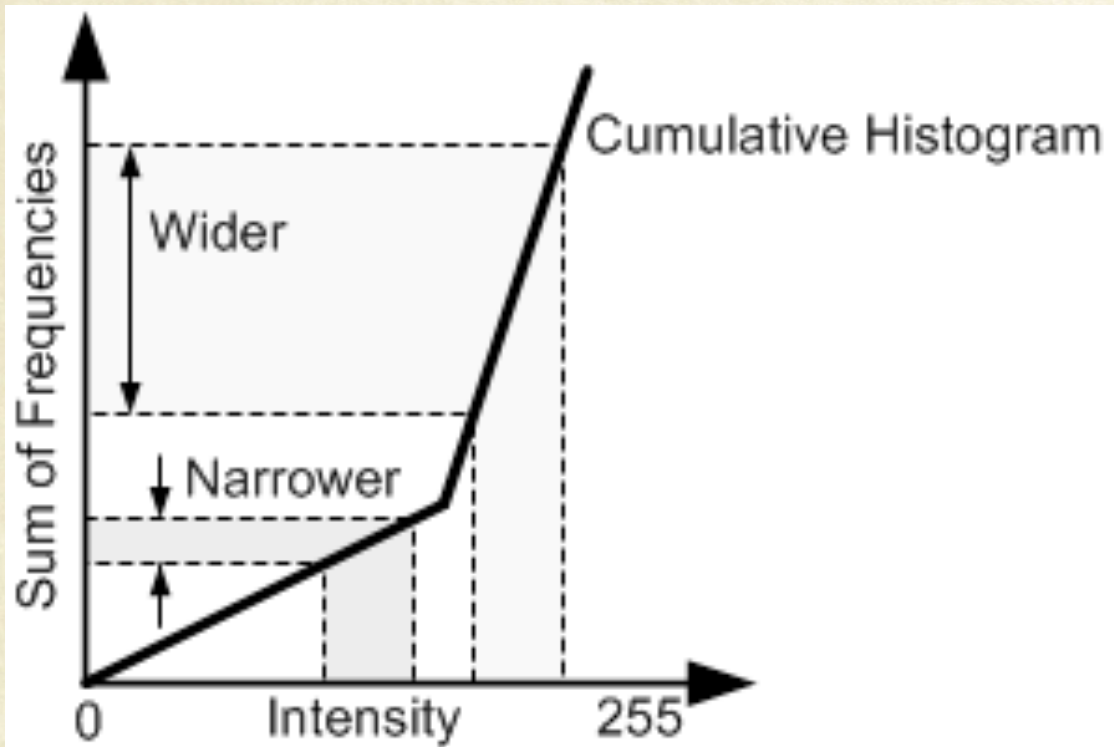
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Histogram Equalization: A Visual Explanation

$$h[i] = \text{constant}, \quad 0 \leq i \leq L - 1$$

$$s = T(r) = (L - 1) \int_0^r p_r(w) dw$$



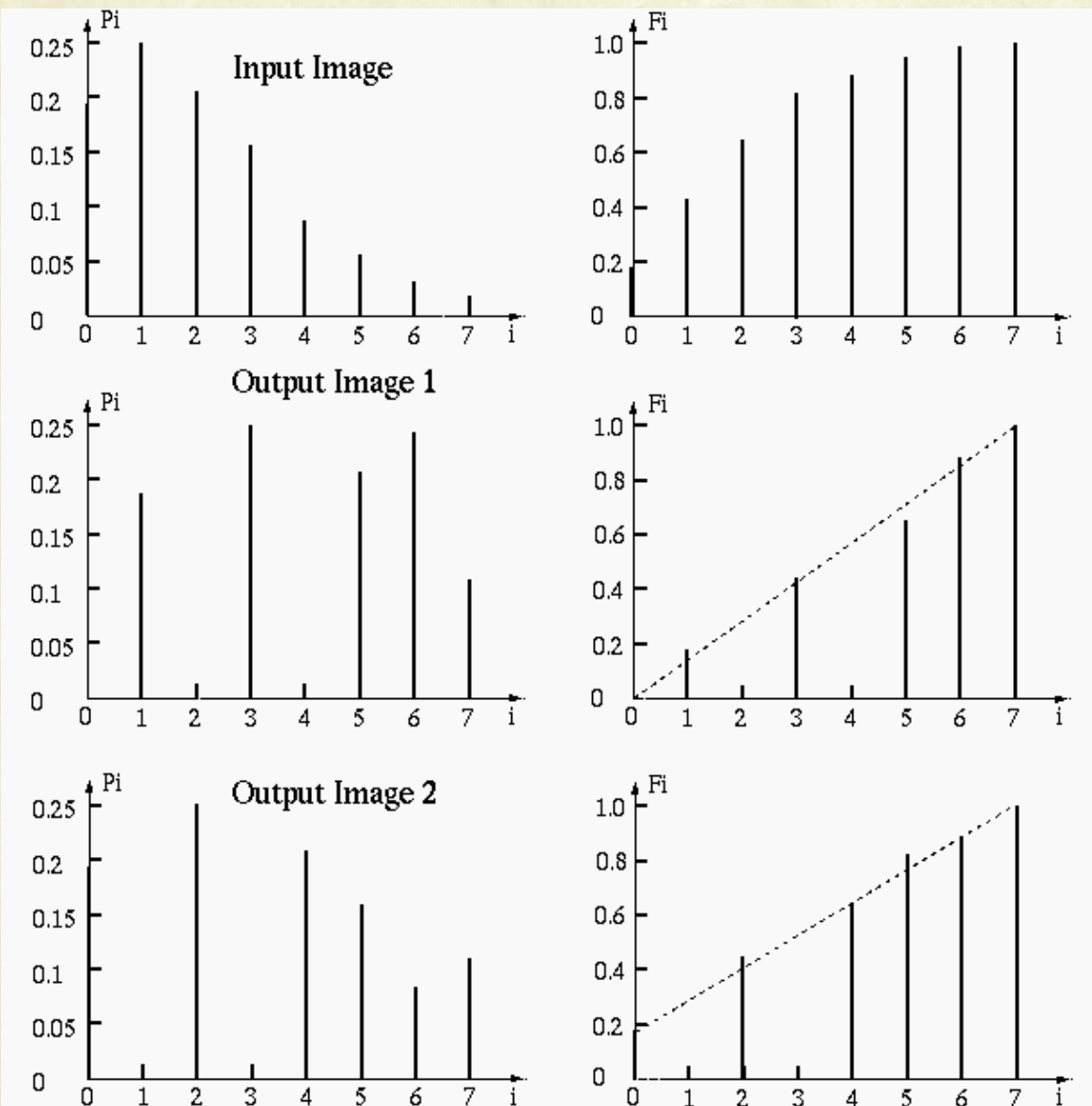
$$s_k = T(r_k) = \text{round} \left((L - 1) \sum_{j=0}^k p_r(r_j) \right)$$

$$s_k = T(r_k) = \text{round} \left((L - 1) \frac{cdf(r_k) - cdf_{min}}{1 - cdf_{min}} \right)$$

$$cdf_{min} = p_r(r_a) \text{ where } r_a = \min\{r_t | p_r(r_t) > 0\}; 0 \leq r_t \leq (L - 1)$$



Histogram Equalization (Default v/s Ver. 2)



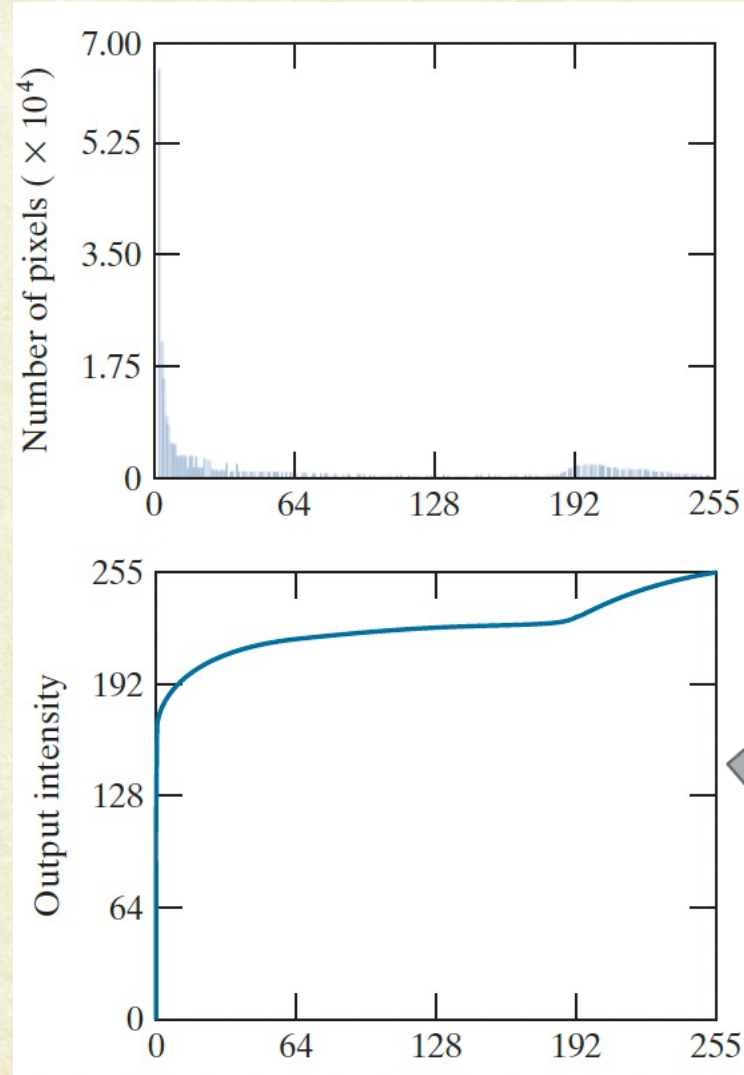
$$s_k = T(r_k) = \text{round} \left((L - 1) \underbrace{\sum_{j=0}^k p_r(r_j)} \right)$$

Ver. 2

$$s_k = T(r_k) = \text{round} \left((L - 1) \frac{cdf(r_k) - cdf_{min}}{1 - cdf_{min}} \right)$$



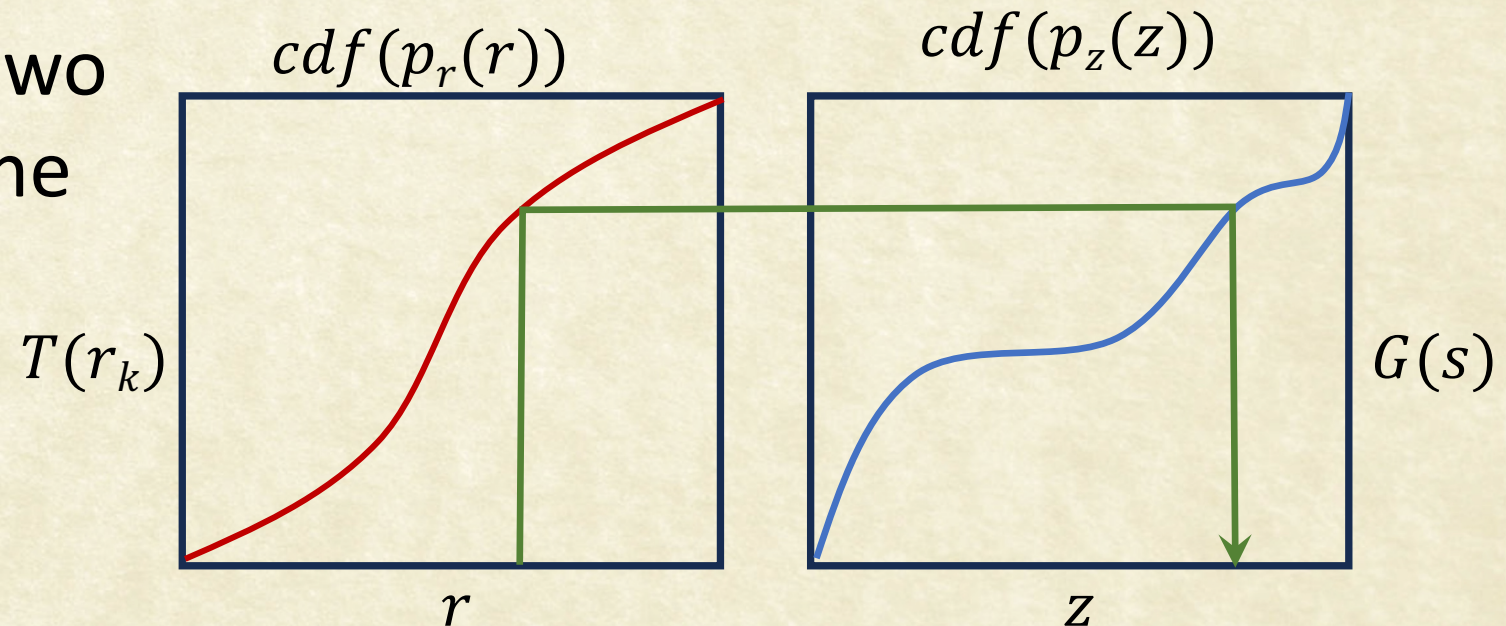
Histogram Equalization





Histogram Specification / Matching

1. Obtain $p_r(r)$ from the input image
2. Compute the function: $s = T(r_k)$ from $p_r(r)$.
3. Obtain $p_z(z)$ from the target image
4. Compute the function $z = G^{-1}(s)$ from $p_z(z)$.
5. Combine the above two functions to obtain the mapping from r to z :
 $z = G^{-1}(T(r_k))$.



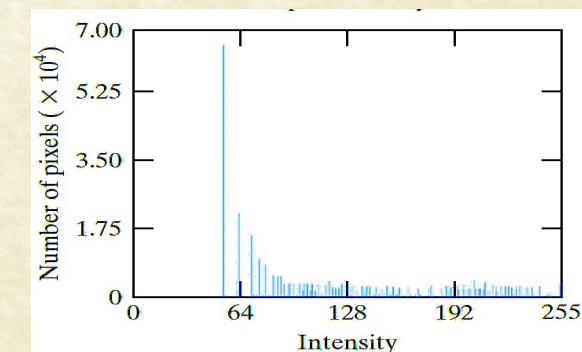
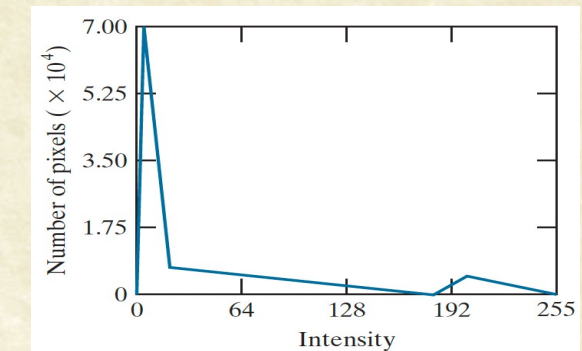
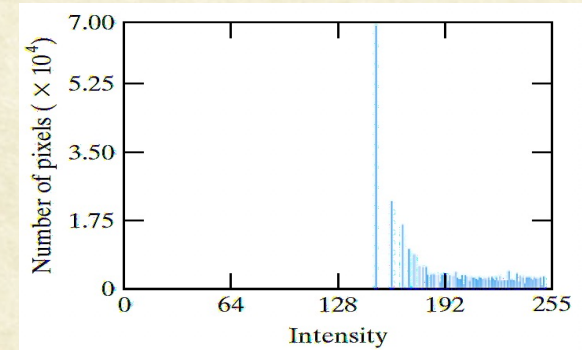
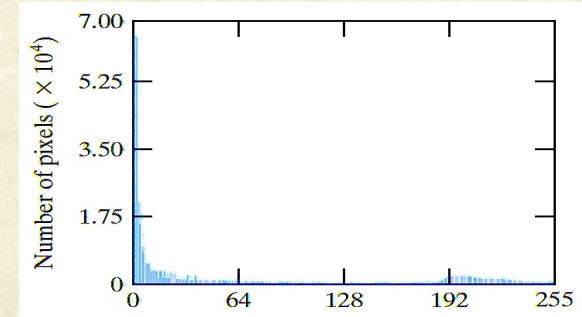


Histogram Specification

Original Image

Histogram Equalized

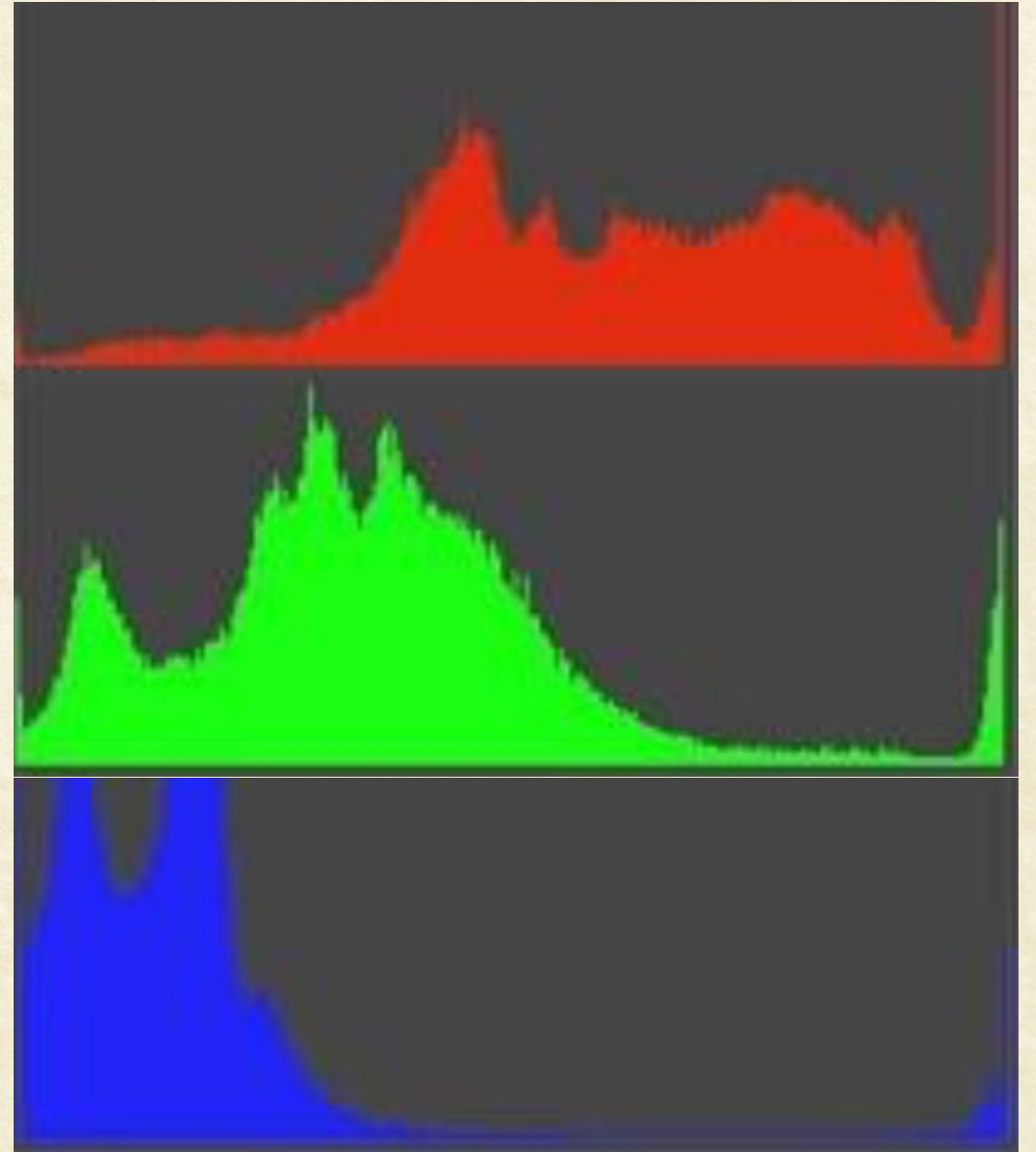
Histogram Specified





Histogram Equalization: Color Image

- Equalize each channel separately





Histogram Equalization: Color Image

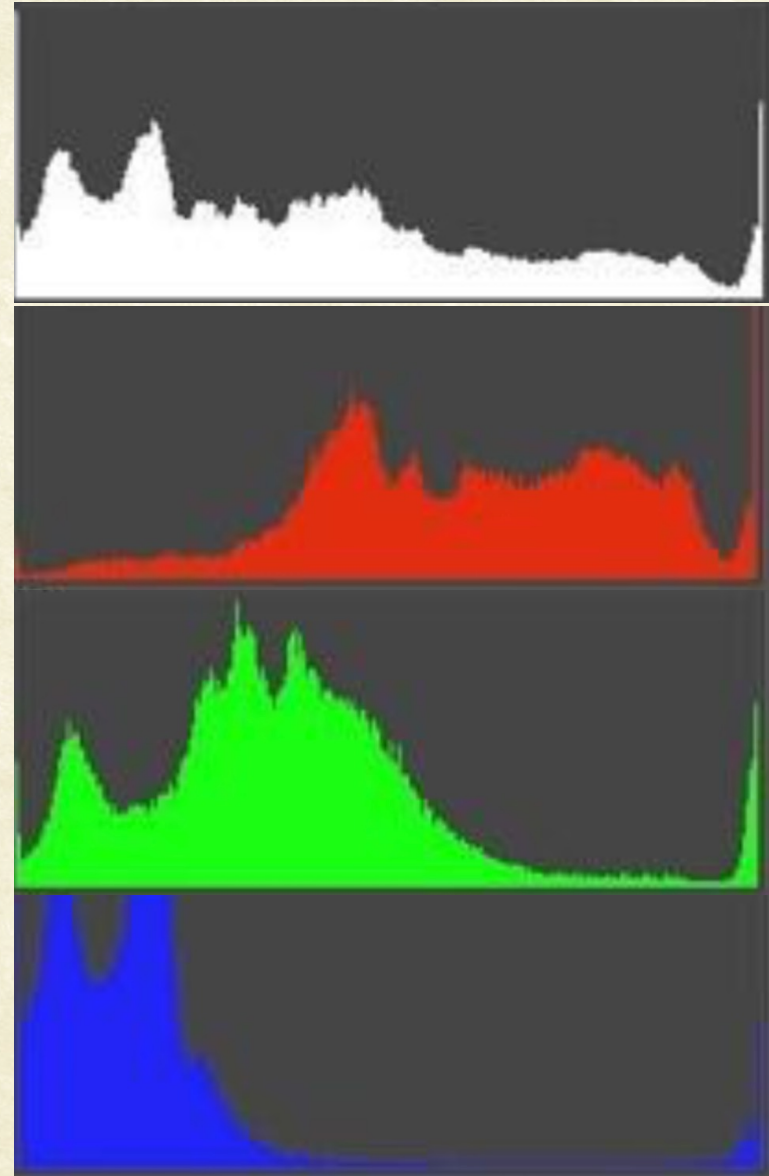
- Equalize each channel separately





Histogram Equalization: Color Image

- Equalize the pooled histogram





Histogram Equalization: Color Image

- Equalize the pooled histogram





Histogram Equalization: Color Image

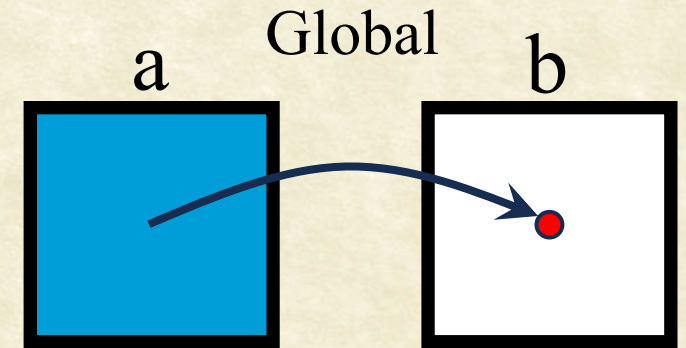
- Equalize the intensity histogram:
 1. RGB $\rightarrow$ HSI;
 2. HistEq(I);
 3. HSI $\rightarrow$ RGB





Histogram : Discussion

- A visualization
- A useful statistical representation of image intensities
 - Not dependent on image size (when normalized)
- Drawbacks
 - No spatial information
 - Intensity-centric
 - Raw (unnormalized form): Image-size dependent
- Equalization:
 - An image 'normalization' approach
 - Improves global contrast, but can also boost noise





References

- ▶ GW Chapter – 3.3 Histogram Processing
- Transformations of Random Variables
 - <http://www.randomservices.org/random/dist/Transformations.html>
 - Section 1 of <http://www.cs.cmu.edu/~minx/transform.pdf>
 - Leibnitz Integration Rule :
https://en.wikipedia.org/wiki/Leibniz_integral_rule#Alternative_derivation
 - [Univariate transformation of a random variable](#)



CS7.404: Digital Image Processing

Monsoon 2025: **II: Spatial vs. Transform Domains**



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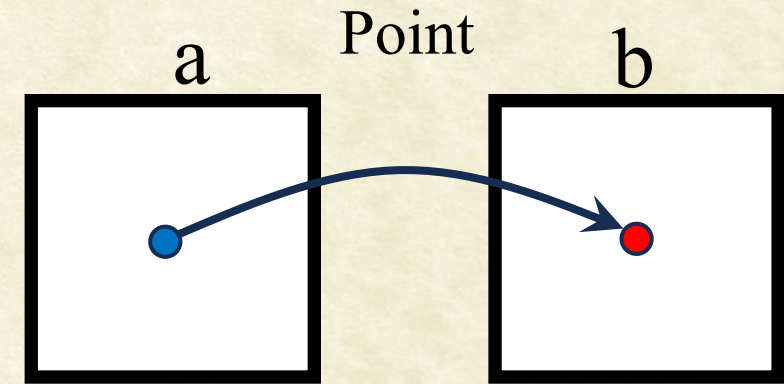
Spatial Domain Processing

Manipulating Pixels Directly in Spatial Domain

Three approaches

I. Point to Point

- Linear Intensity Transforms
 - e.g., Negative
- Non-linear Transforms
 - e.g., Logarithm
- Histogram





Spatial Domain Processing

Manipulating Pixels Directly in Spatial Domain

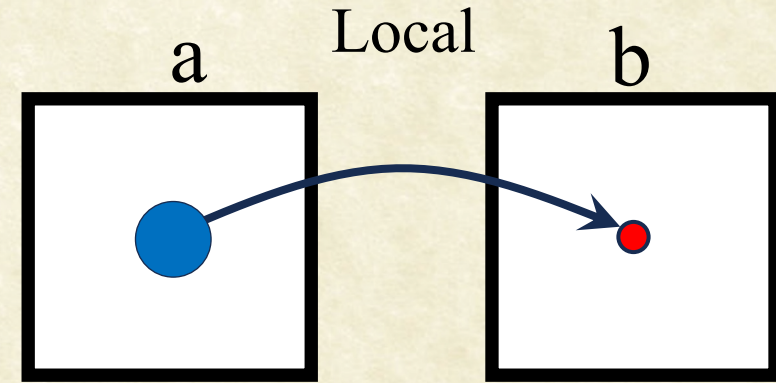
Three approaches

2. Neighborhood to Point

48	219	168	145	244	188	120	58
49	218	87	94	34	104	118	72
174	151	74	179	108	118	127	124
77	127	87	139	163	143	131	117
138	219	146	108	132	129	122	109
38	208	174	168	184	148	121	101
178	164	79	158	191	137	116	96
96	209	214	203	194	150	113	74

Current pixel

3 x 5 neighbourhood



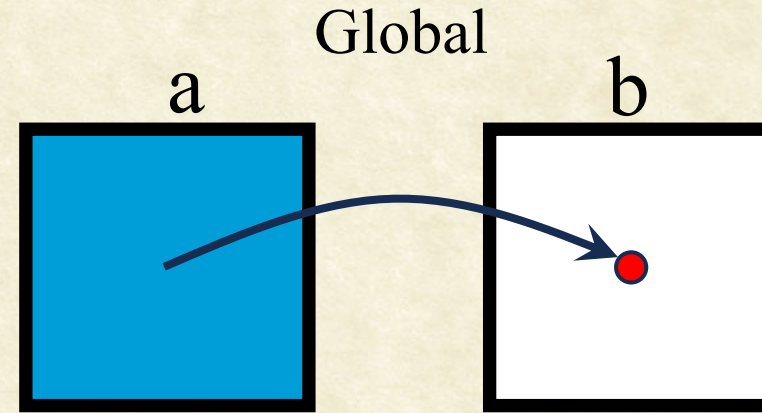


Spatial Domain Processing

Manipulating Pixels Directly in Spatial Domain

Three approaches

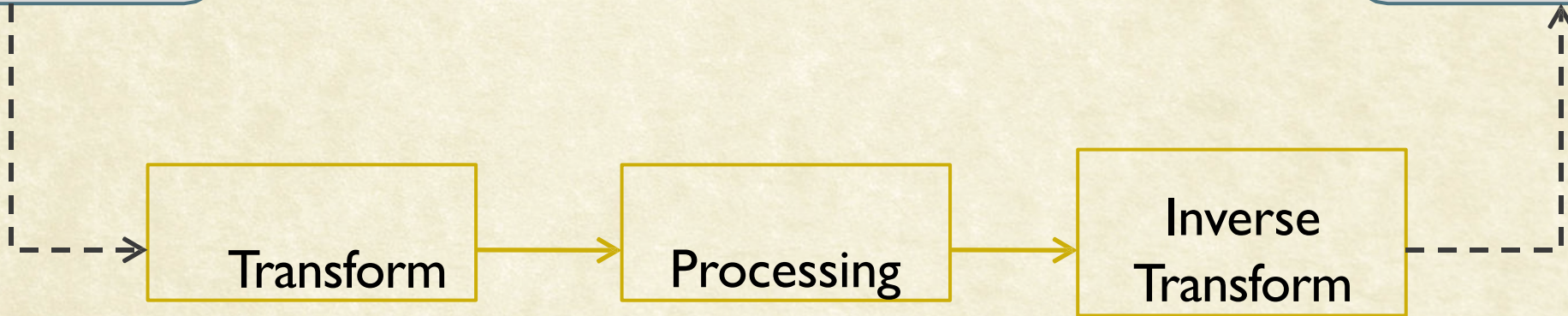
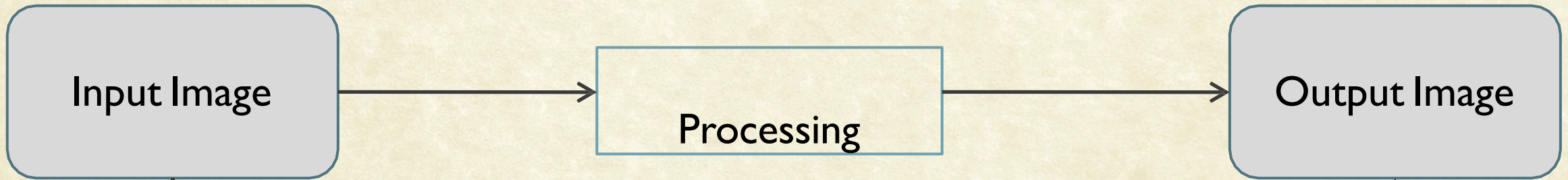
3. Global to Point





Spatial vs. Transform Domain Processing

Spatial Domain



Transform Domain



Spatial vs. Transform Domain Processing



Bandhani / Bandhej



Tie & Dye



Spatial vs. Transform Domain Processing

Transform (Tie)



Process (Dye)

Inverse Transform (Untie)



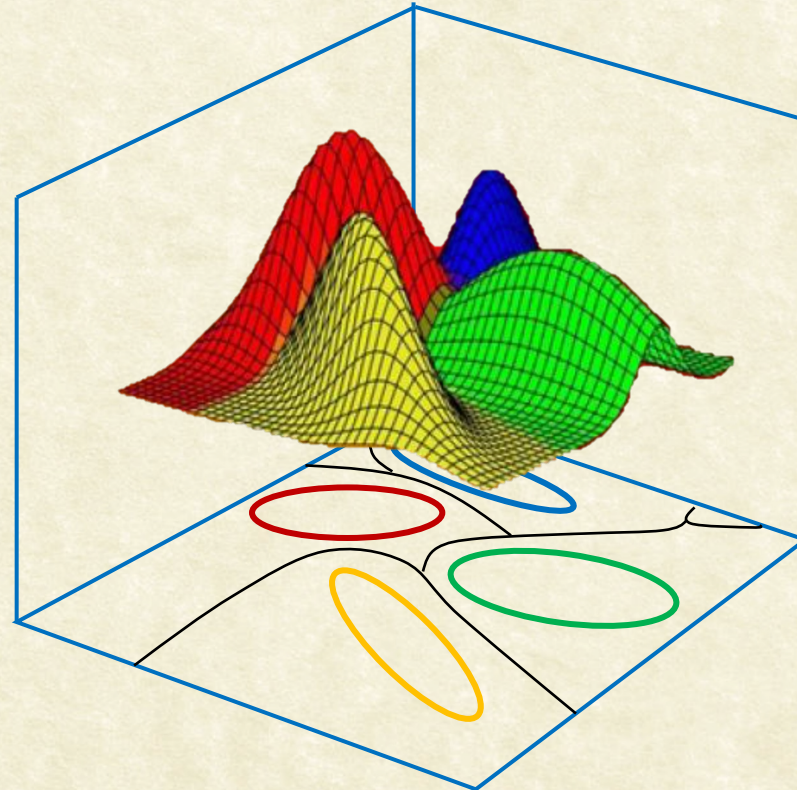


Questions?



CS7.404: Digital Image Processing

Monsoon 2025: **III**: Spatial Domain Processing



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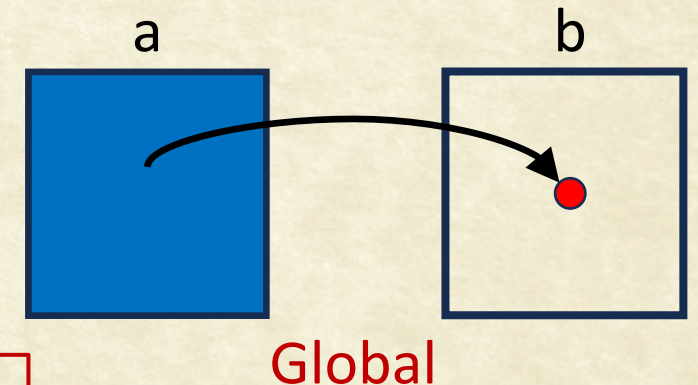
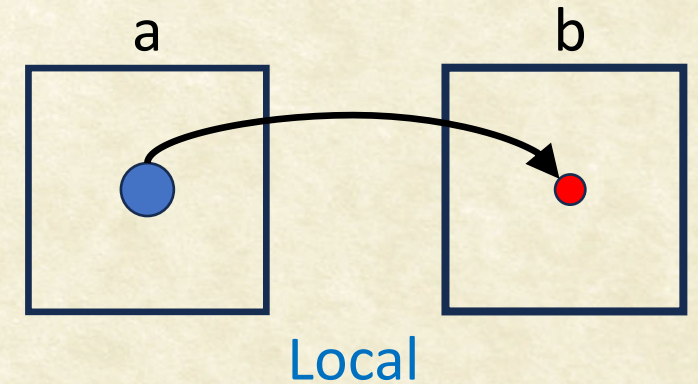


Spatial Domain Processing

► Manipulating Pixels Directly in Spatial Domain

3 approaches

1. Point to Point
2. Neighbourhood to Point
3. Global to Point



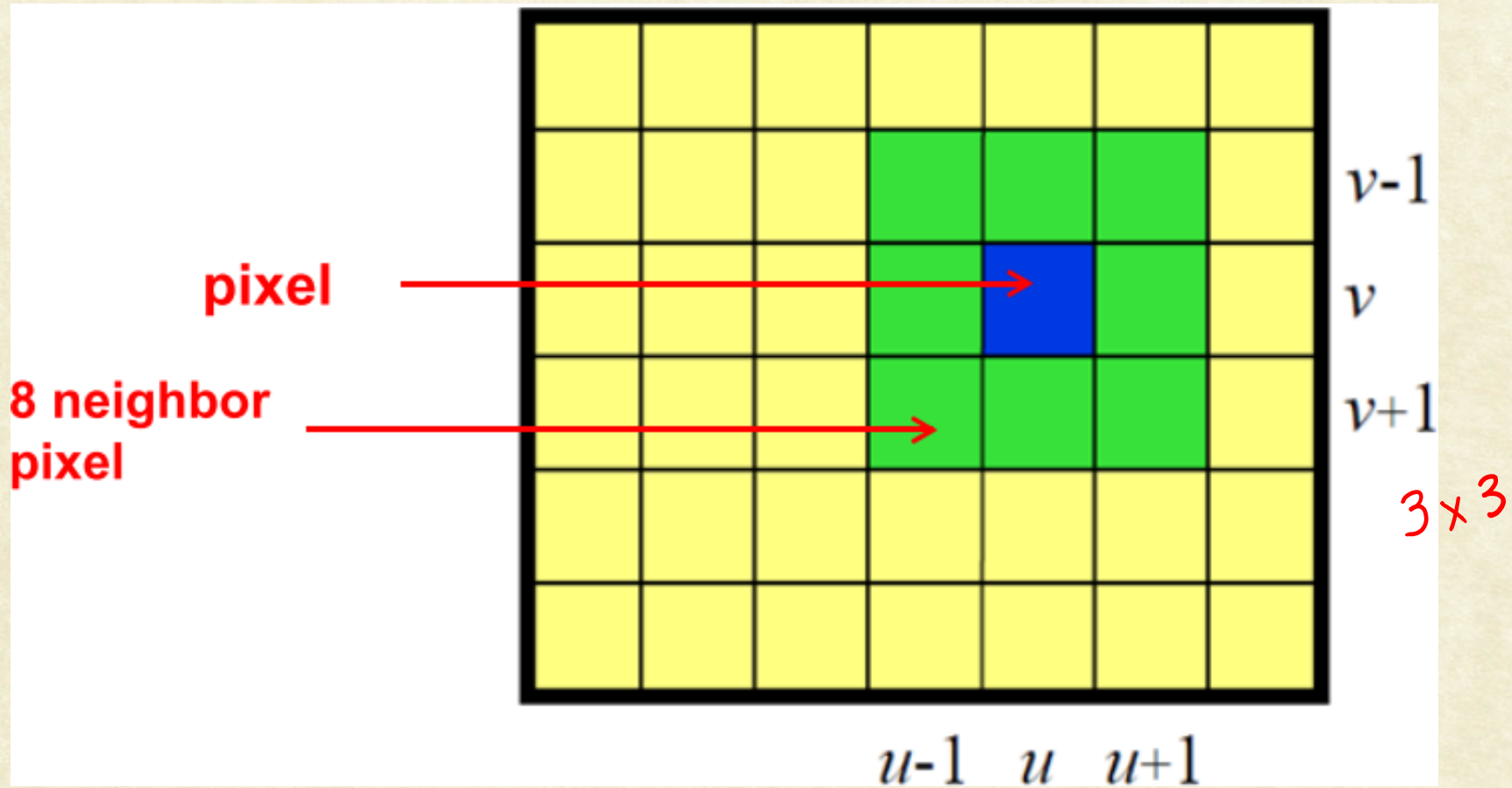
48	219	168	145	244	188	120	58
49	218	87	94	34	104	118	72
174	151	74	179	108	118	127	124
77	127	87	139	163	143	131	117
138	219	146	108	132	129	122	109
38	208	174	168	184	148	121	101
178	164	79	158	191	137	116	96
96	209	214	203	194	150	113	74

Current pixel

3 x 5 neighbourhood

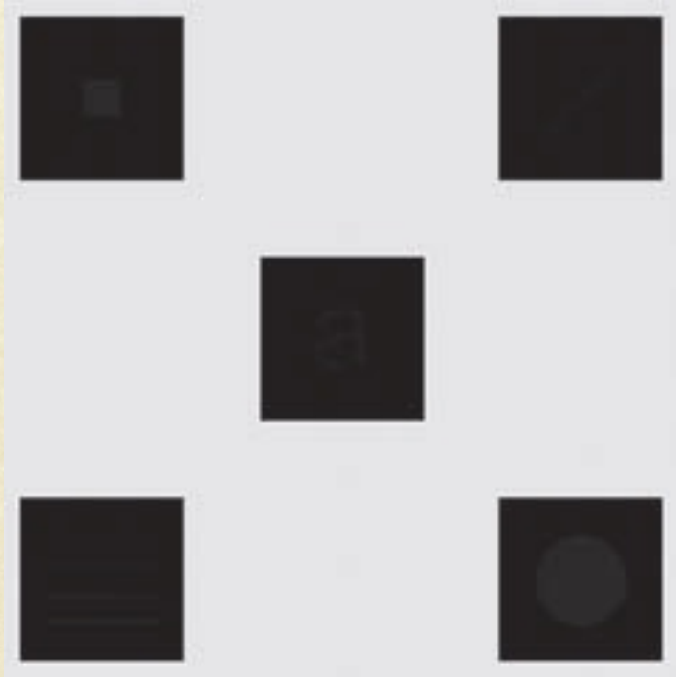


Pixel Neighborhood

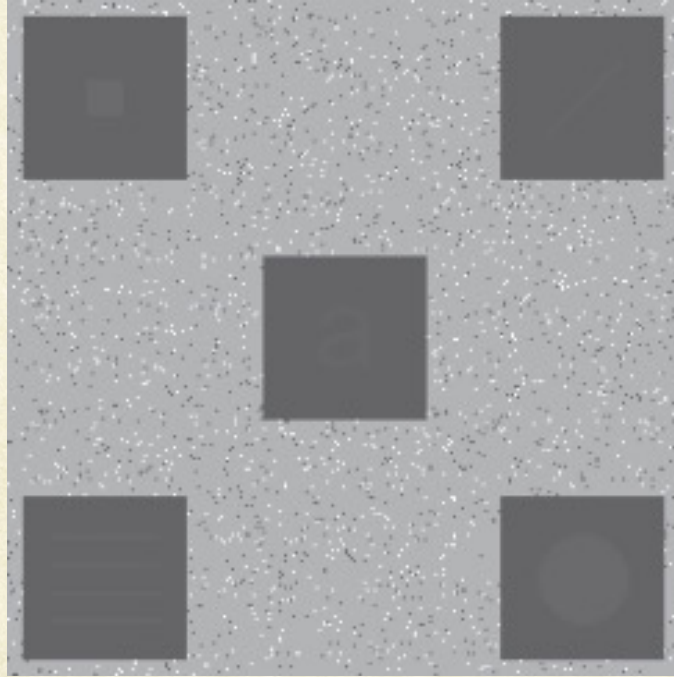




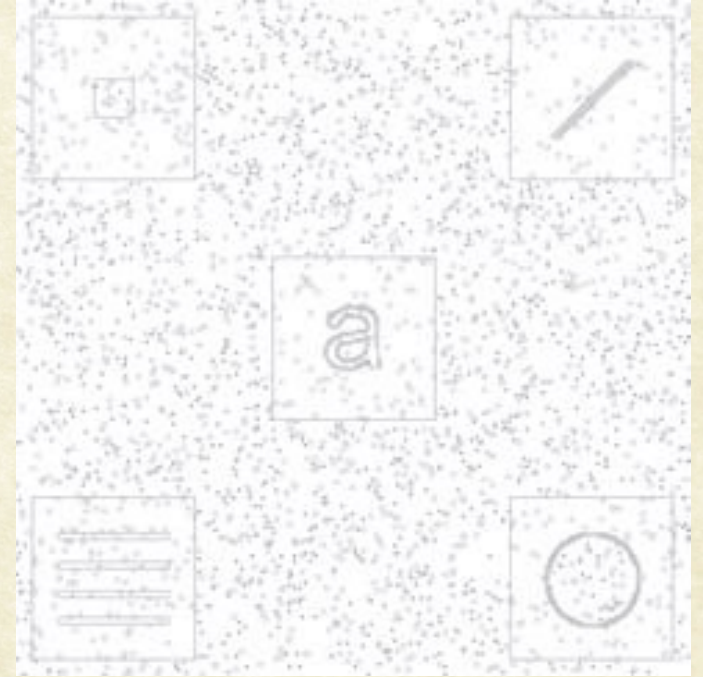
Local/Adaptive Histogram Equalization



(a) Original image



(b) Result of global
histogram equalization

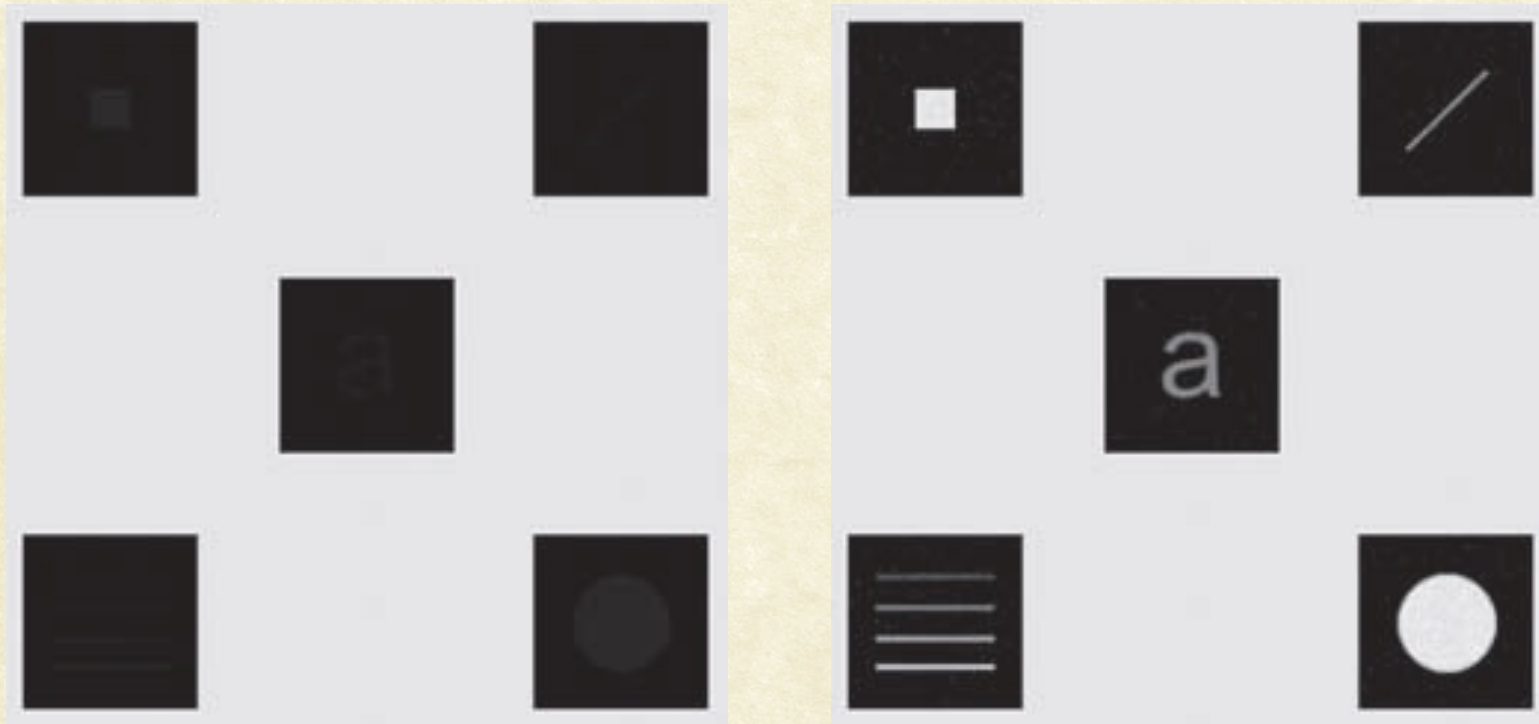


(c) Result of local
histogram equalization



Local Histogram Statistics for Enhancement

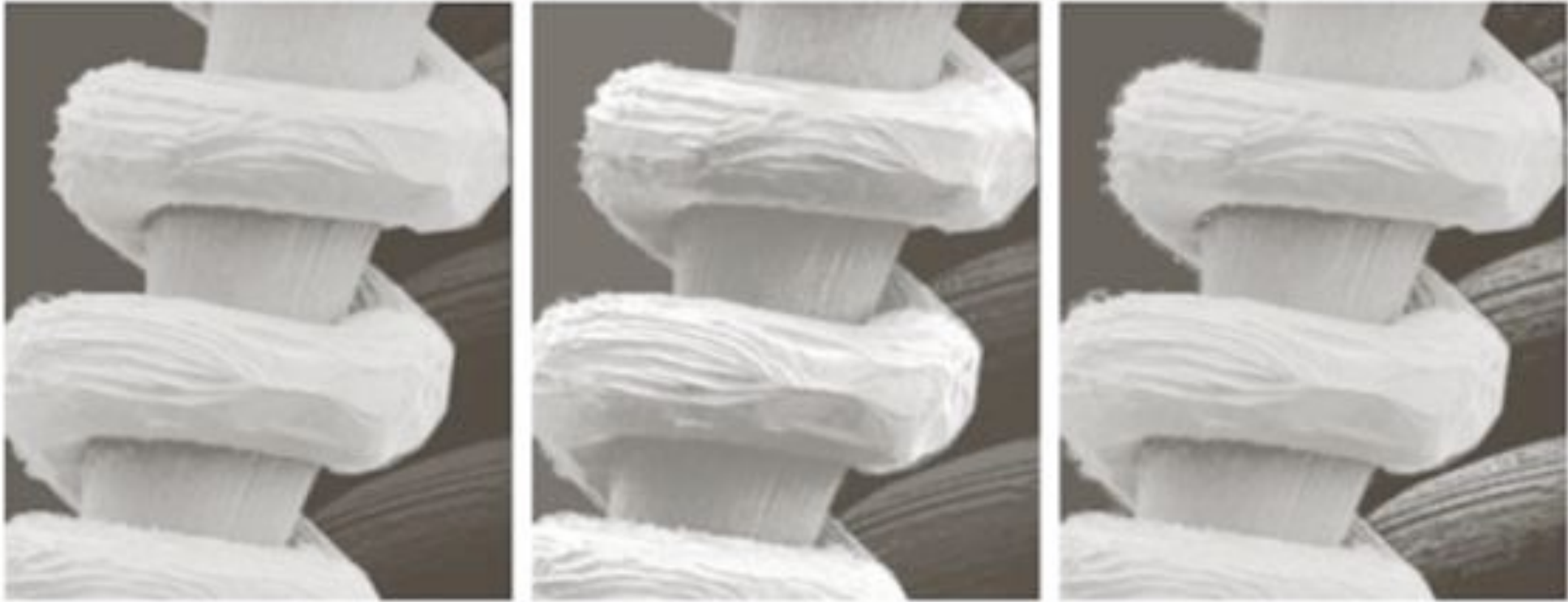
$$g(x, y) = \begin{cases} Cf(x, y) & \text{if } k_0 m_G \leq m_{S_{xy}} \leq k_1 m_G \text{ AND } k_2 \sigma_G \leq \sigma_{S_{xy}} \leq k_3 \sigma_G \\ f(x, y) & \text{otherwise} \end{cases}$$



- For Self Reading: Contrast-Limited Adaptive Histogram Equalization (CLAHE)



Image Enhancement Using Histogram Statistics



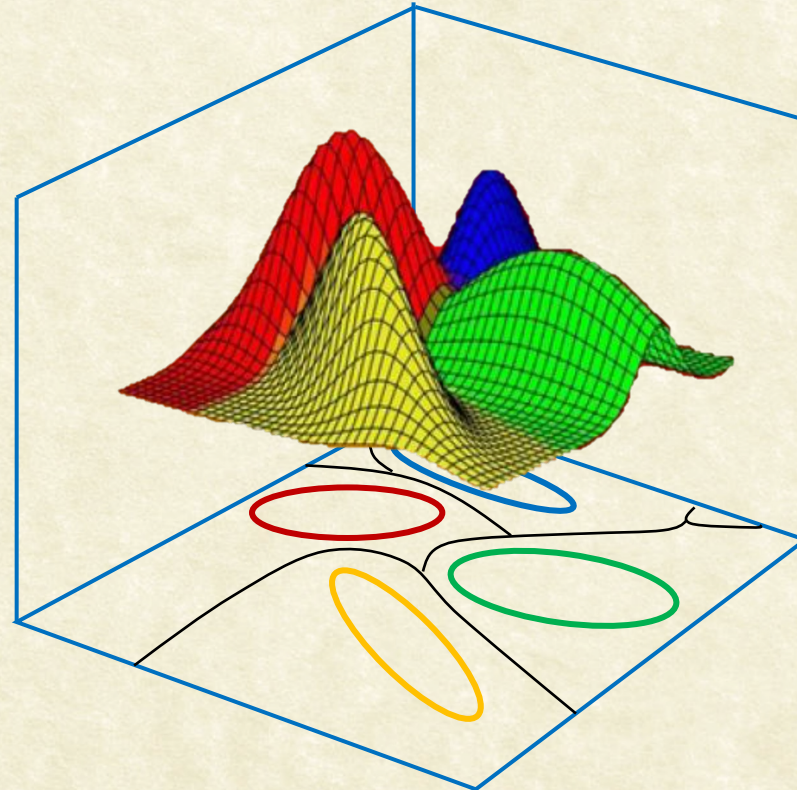
a b c

FIGURE 3.27 (a) SEM image of a tungsten filament magnified approximately $130\times$. (b) Result of global histogram equalization. (c) Image enhanced using local histogram statistics. (Original image courtesy of Mr. Michael Shaffer, Department of Geological Sciences, University of Oregon, Eugene.)



CS7.404: Digital Image Processing

Monsoon 2025 : **IV:** Spatial Filtering



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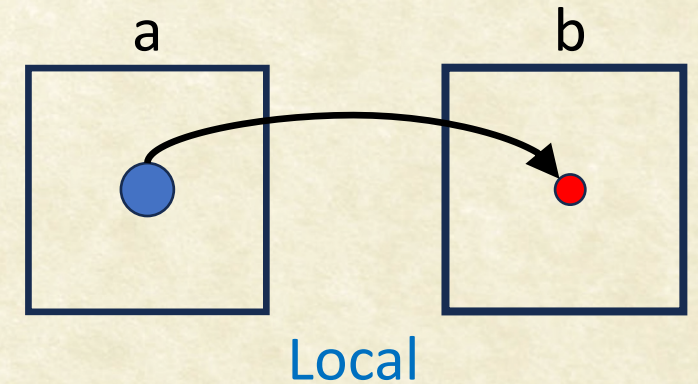


Spatial Domain Processing

► Manipulating Pixels Directly in Spatial Domain

3 approaches

1. Point to Point
2. Neighbourhood to Point





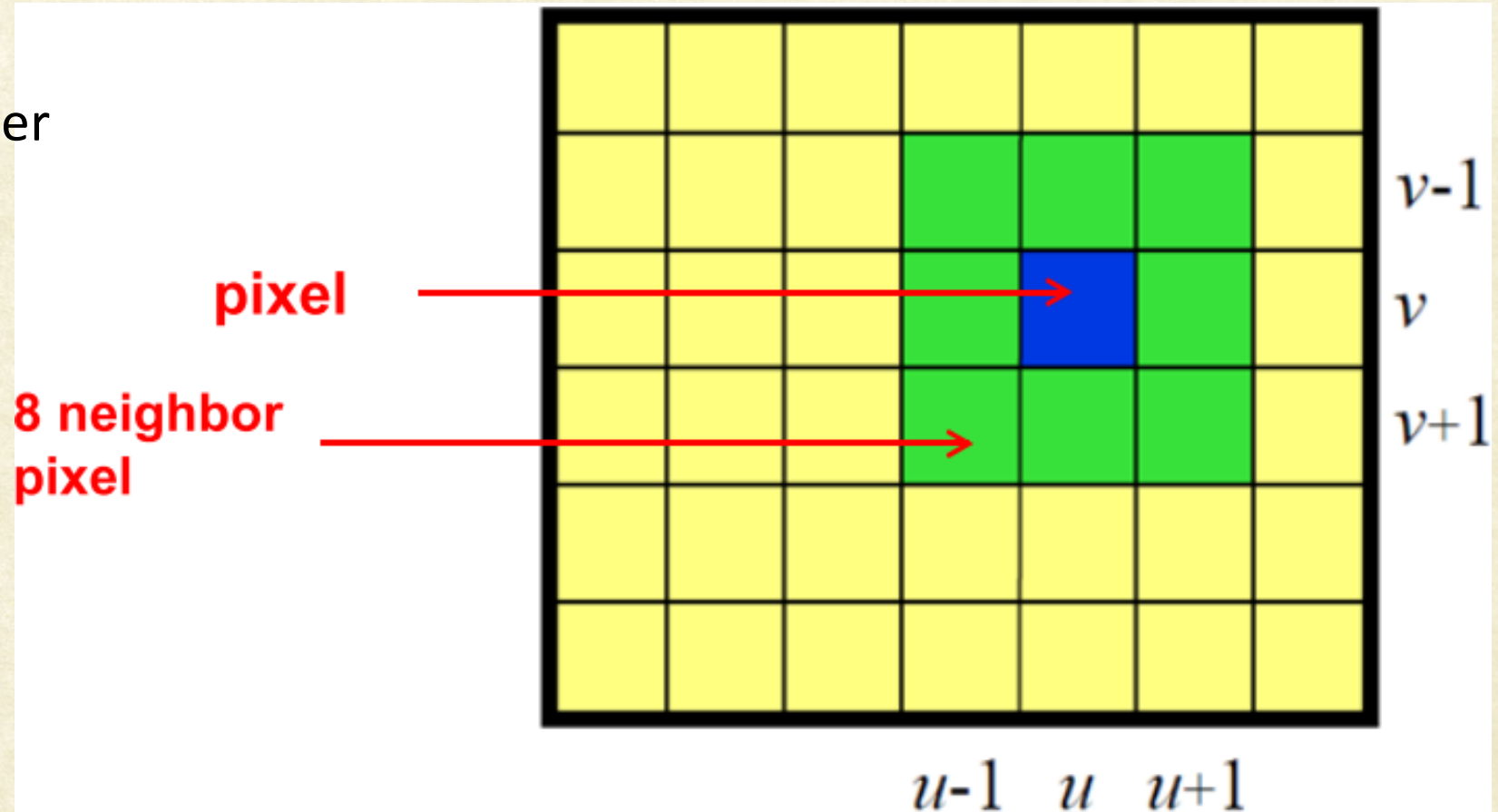
Spatial Domain Filtering

2 components of a spatial filter

- neighborhood
- a predefined operation

Filter aka

- Kernel
- Spatial Mask
- Template
- Window





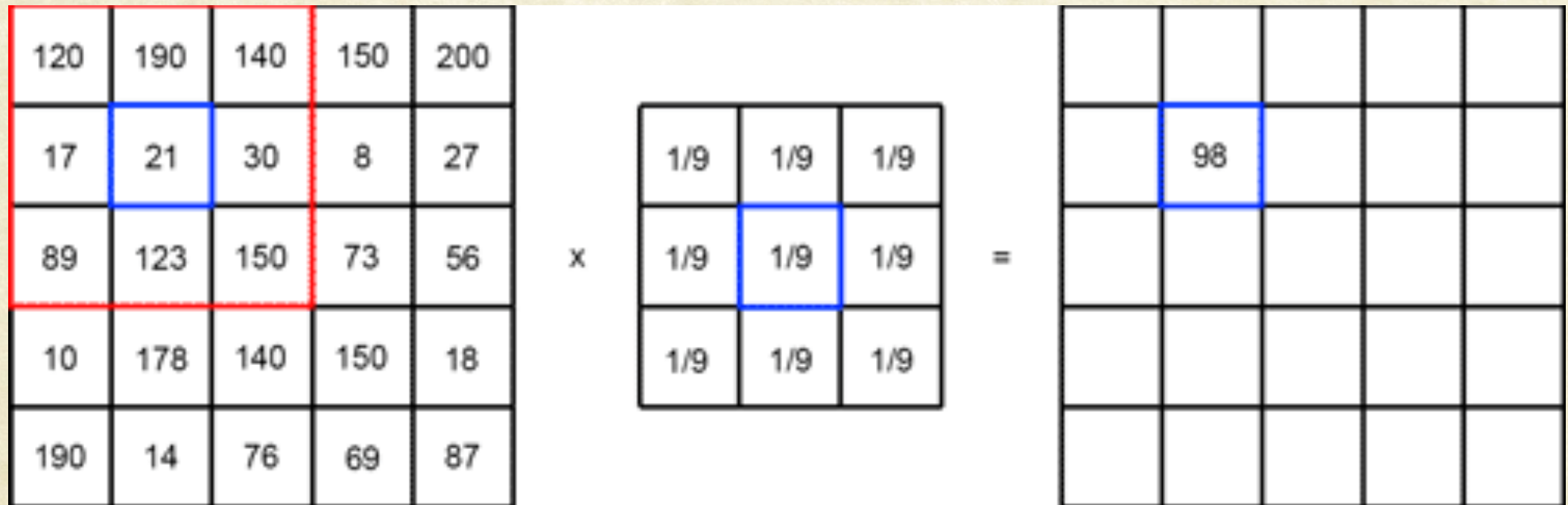
Mean/Average Filter (Smoothing)

$M = 3$

For each valid location $[x,y]$ in S

$a \leftarrow$ Average of intensities in a $M \times M$ neighborhood centered on $[x,y]$

$D[x,y] = \text{round}(a)$





$$\frac{1}{9} * \begin{matrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{matrix}$$





Mean / Average Filter

Note: Coefficients sum to 1

I

H

→ 3 x 3

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

Weight Mask /
Kernel /
Filter

$$I'(u, v) \leftarrow \frac{1}{9} \cdot \sum_{j=-1}^1 \sum_{i=-1}^1 I(u + i, v + j)$$

$$I'(u, v) \leftarrow \sum_{j=-1}^1 \sum_{i=-1}^1 I(u + i, v + j) \cdot \boxed{H(i, j)}$$



Effect of Mask Size

Original Image



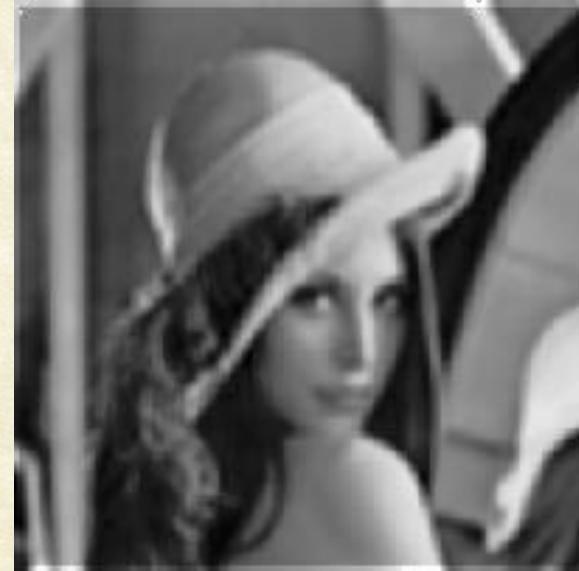
[3x3]



[5x5]



[7x7]





Square averaging filter

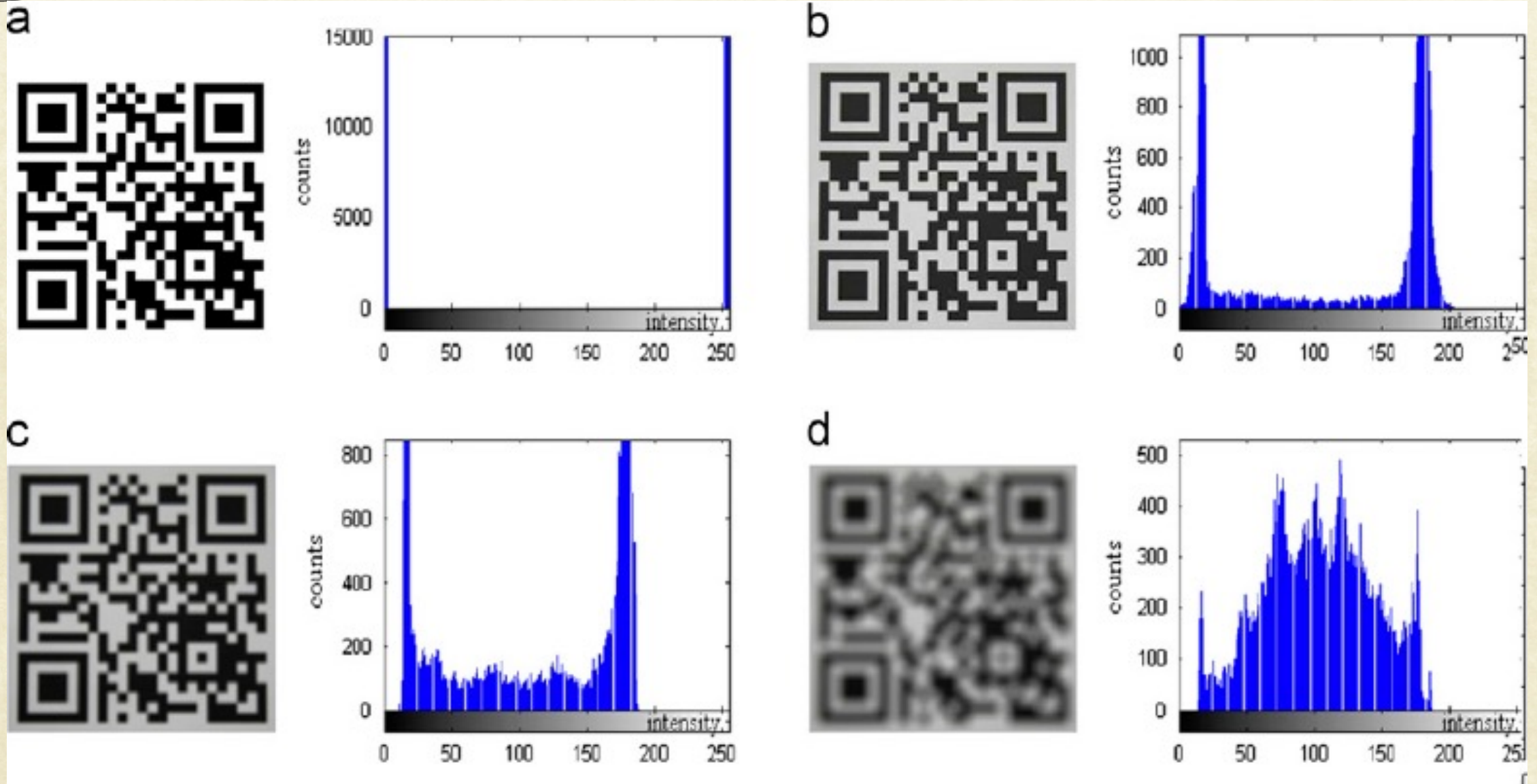
FIGURE 3.33 (a) Original image, of size 500×500 pixels. (b)–(f) Results of smoothing with square averaging filter masks of sizes $m = 3, 5, 9, 15$, and 35 , respectively. The black squares at the top are of sizes $3, 5, 9, 15, 25, 35, 45$, and 55 pixels, respectively; their borders are 25 pixels apart. The letters at the bottom range in size from 10 to 24 points, in increments of 2 points; the large letter at the top is 60 points. The vertical bars are 5 pixels wide and 100 pixels high; their separation is 20 pixels. The diameter of the circles is 25 pixels, and their borders are 15 pixels apart; their intensity levels range from 0% to 100% black in increments of 20% . The background of the image is 10% black. The noisy rectangles are of size 50×120 pixels.

a b
c d
e f





Averaging – a histogram perspective





Repeated Averaging Using Same Filter



Before



After



After repeated
averaging

NOTE: Can get the effect of larger filters by smoothing repeatedly with smaller filters



Weighted Averaging

$$\frac{1}{9} \times$$

1	1	1
1	1	1
1	1	1

Standard average

$$\frac{1}{16} \times$$

1	2	1
2	4	2
1	2	1

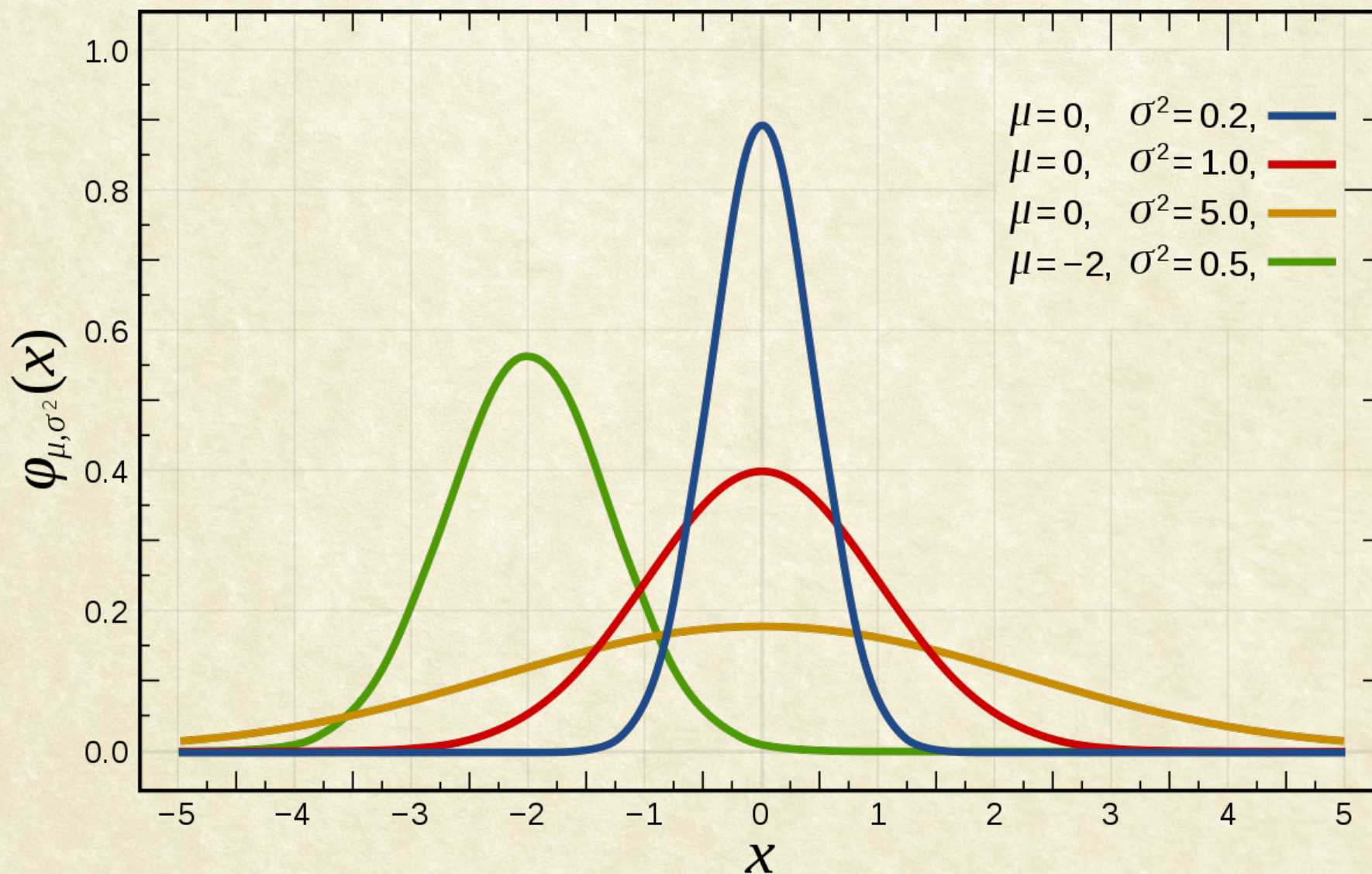
Weighted Average

$$I'(u, v) \leftarrow \frac{\sum_{j=-a}^a \sum_{i=-b}^b I(u+i, v+j) \cdot H(i, j)}{\sum_{j=-a}^a \sum_{i=-b}^b H(i, j)}$$



Gaussian Function (1-D)

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

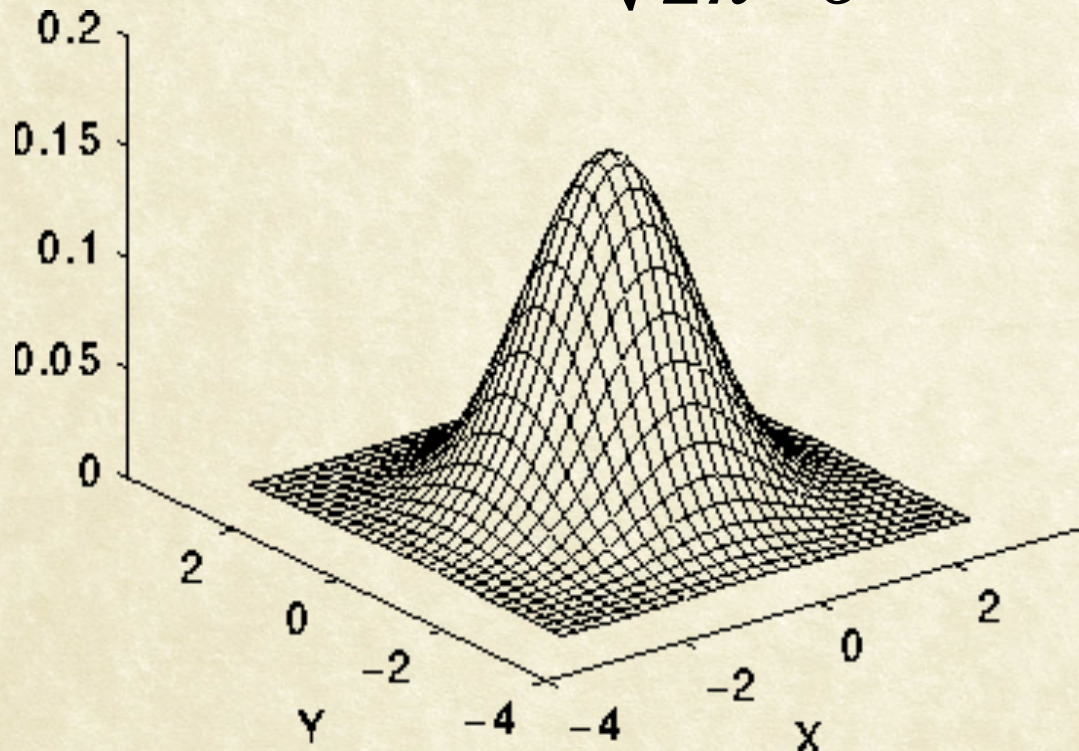




Gaussian Smoothing

- ▶ Mask weights are samples of a zero-mean 2-D Gaussian

$$G(x, y) = \frac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$



$$\frac{1}{265}$$

1	4	6	4	1
4	16	26	16	4
6	26	43	26	6
4	16	26	16	4
1	4	6	4	1

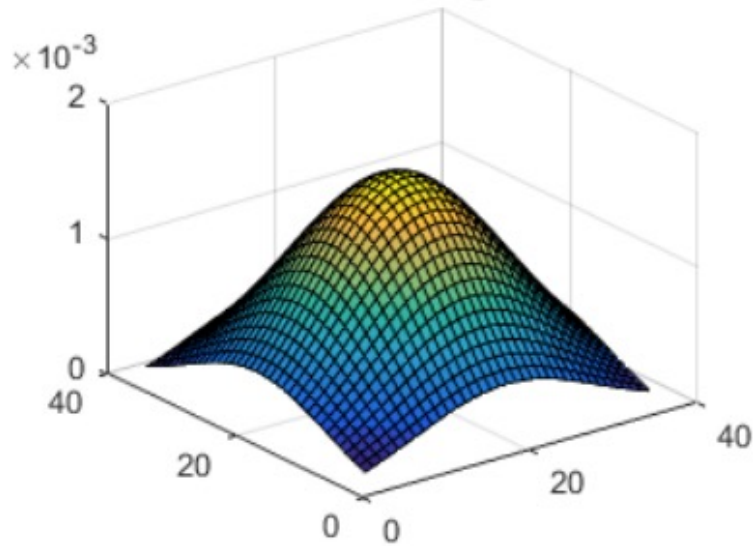
5×5 Gaussian filter, $\sigma = 1$



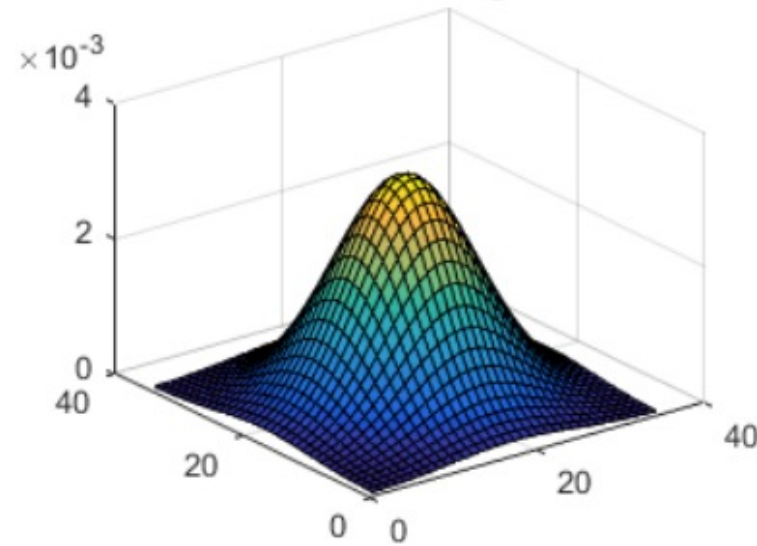
Gaussian Smoothing – Effect of sigma

$$G(x, y) = \frac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$

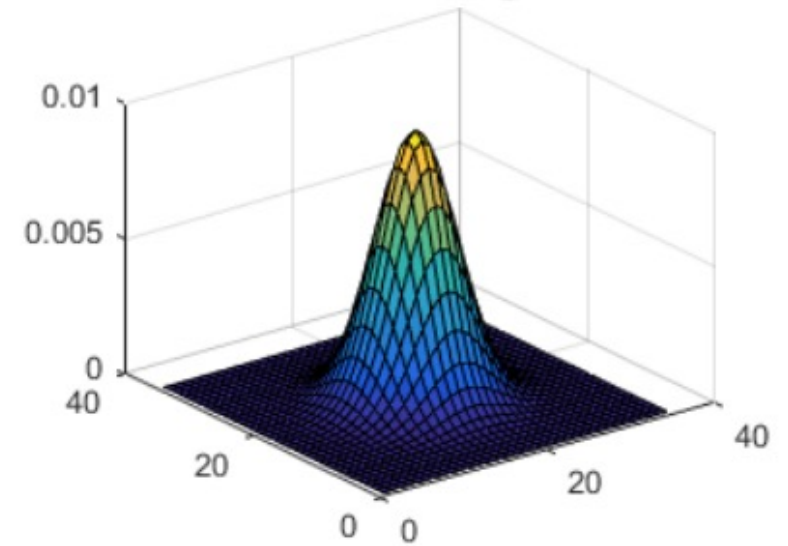
filter size = 35, sigma = 11



filter size = 35, sigma = 7



filter size = 35, sigma = 2





Gaussian Smoothing – Effect of sigma



Original Image
(Sigma 0)

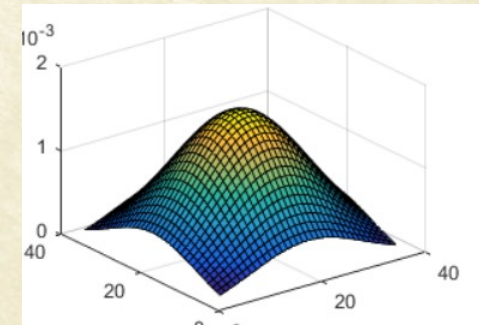
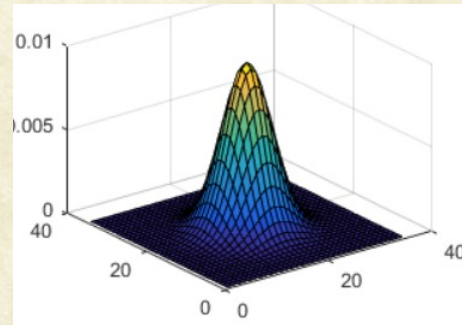


Gaussian Blur
(Sigma 0.7)



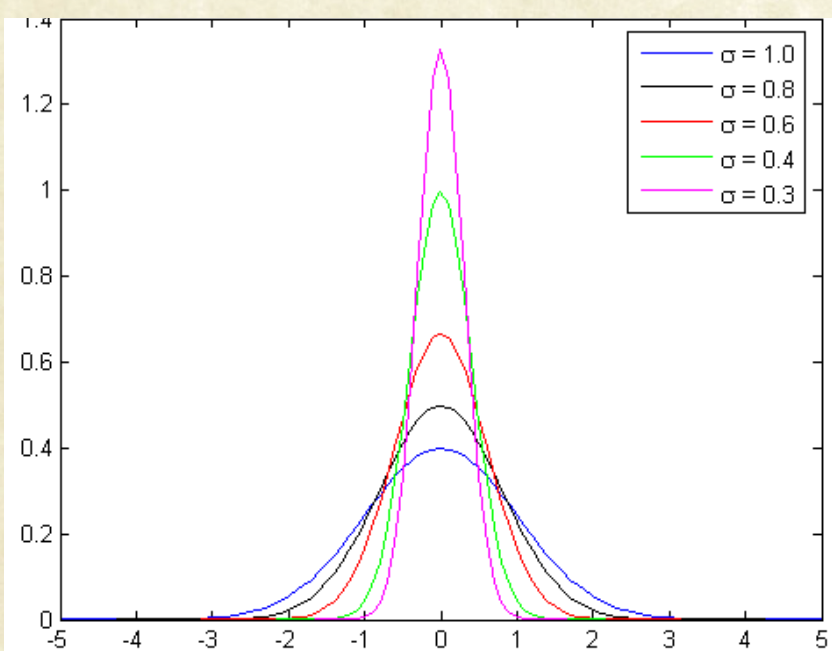
Gaussian Blur
(Sigma 2.8)

$$G(x, y) = \frac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$

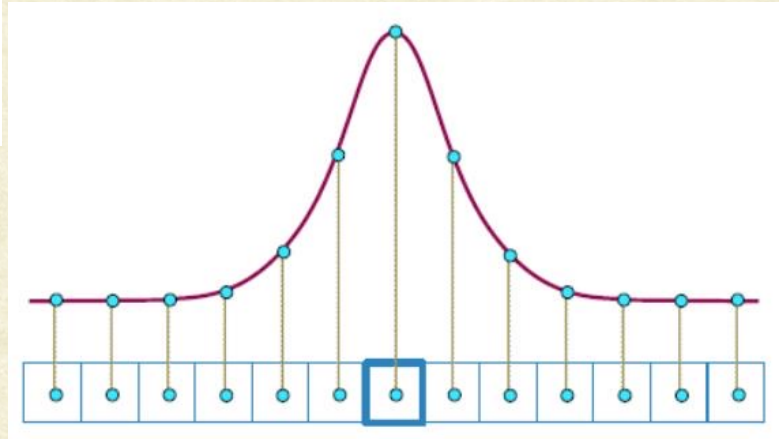
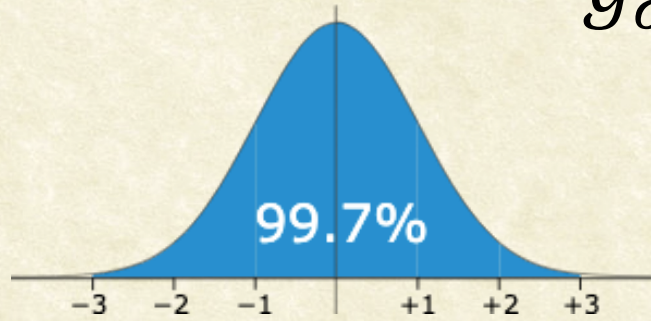




Computing Gaussian filter coefficients

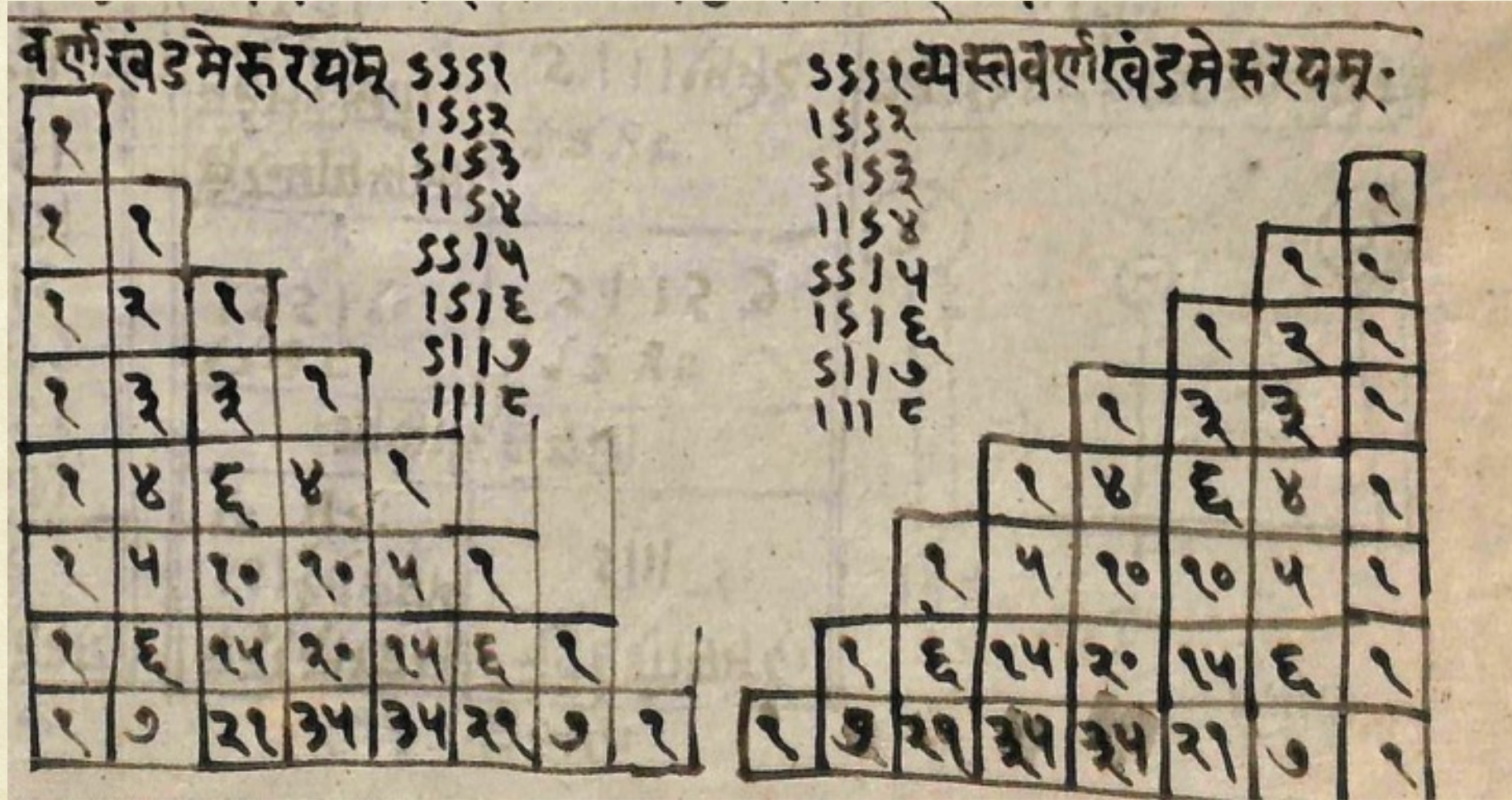


$$g_{\sigma}(x) = \frac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\left(\frac{x^2}{2\sigma^2}\right)}$$





Computing Gaussian filter coefficients



Meru Prastaara, derived from Pingala's formulae (2 BCE),
Manuscript from Raghunath Temple Library, Jammu

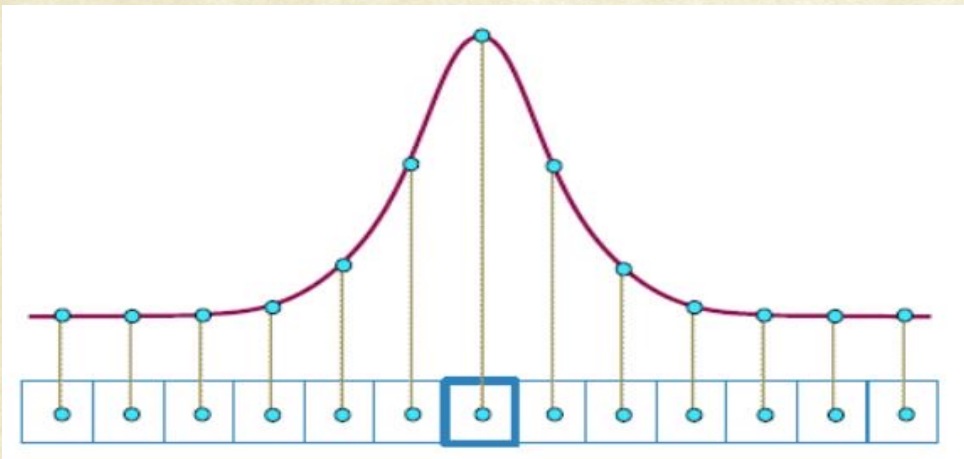


Computing Gaussian filter coefficients

Index N	Coefficients	Sum= 2^N
0	1	1
1	1 1	2
2	1 2 1	4
3	1 3 3 1	8
4	1 4 6 4 1	16
5	1 5 10 10 5 1	32
6	1 6 15 20 15 6 1	64
7	1 7 21 35 35 21 7 1	128
8	1 8 28 56 70 56 28 8 1	256
9	1 9 36 84 126 126 84 36 9 1	512
10	1 10 45 120 210 252 210 120 45 10 1	1024

e.g., $s = 7 \times 7$

$$\frac{1}{64} \times \begin{matrix} 1 & 6 & 15 & 20 & 15 & 6 & 1 \end{matrix}$$



$$g_{\sigma}(x) = \frac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\left(\frac{x^2}{2\sigma^2}\right)}$$



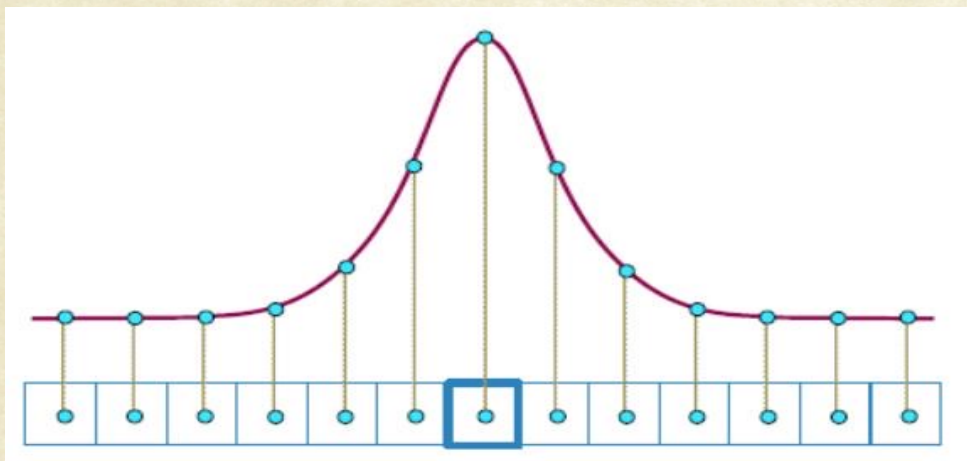
Computing Gaussian filter coefficients

Index N	Coefficients	Sum=2 ^N
0	1	1
1	1 1	2
2	1 2 1	4
3	1 3 3 1	8
4	1 4 6 4 1	16
5	1 5 10 10 5 1	32
6	1 6 15 20 15 6 1	64
7	1 7 21 35 35 21 7 1	128
8	1 8 28 56 70 56 28 8 1	256
9	1 9 36 84 126 126 84 36 9 1	512
10	1 10 45 120 210 252 210 120 45 10 1	1024

e.g., $s = 7 \times 7$

$$\frac{1}{4096} \times$$

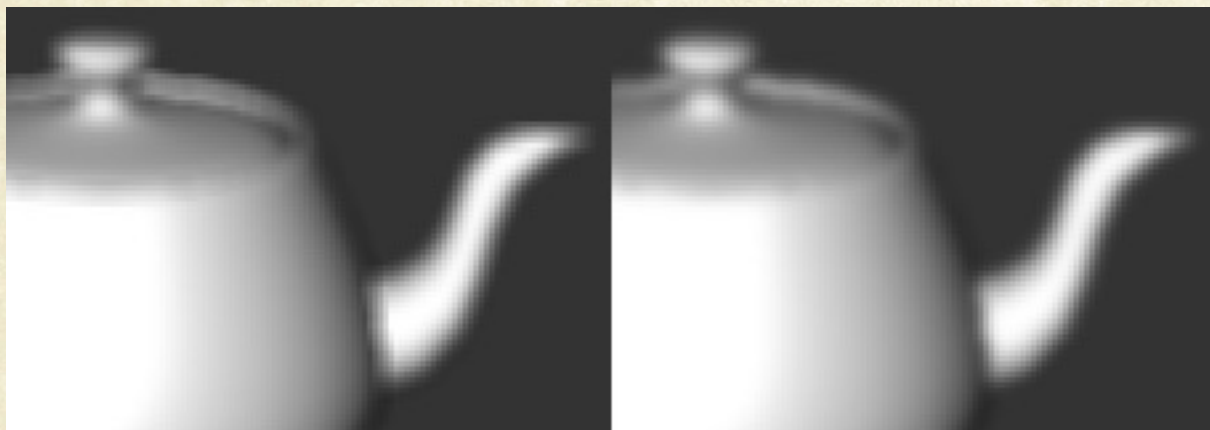
1	6	15	20	15	6	1
6	36	90	120	90	36	6
15	90	225	300	225	90	15
20	120	300	400	300	120	20
15	90	225	300	225	90	15
6	36	90	120	90	36	6
1	6	15	20	15	6	1



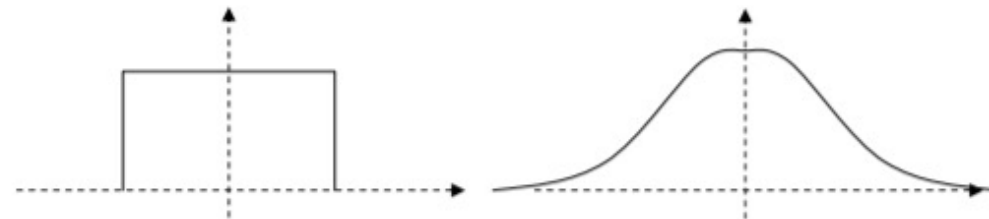
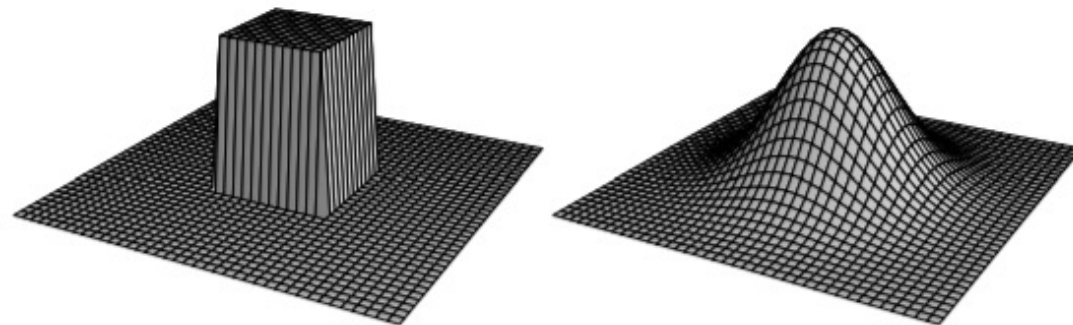
$$g_{\sigma}(x) = \frac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\left(\frac{x^2}{2\sigma^2}\right)}$$



Averaging vs. Gaussian Filter



Smother intensity transitions



0	0	0	0	0
0	1	1	1	0
0	1	1	1	0
0	1	1	1	0
0	0	0	0	0

(a)

0	1	2	1	0
1	3	5	3	1
2	5	9	5	2
1	3	5	3	1
0	1	2	1	0

(b)



Questions?